

Merged research report for the Surreal project

Omnific-Preserving Automorphisms

Convex-scale stabilizers, definable constants,
nondefinable monomials, and algebraic-parameter rigidity

Built from four manuscripts dated September 23, 2026

Abstract

Which symmetries of a Hahn field preserve its omnific-type integer part, and what does that integer part remember? Let k have characteristic zero, let $\Gamma \neq 0$ be a divisible ordered abelian group, write C_δ for the principal convex subgroup generated by $\delta > 0$, and let $\mathfrak{o} \subseteq k$ be a unital subring. A strong coefficient-fixing 1-automorphism σ of $k((t^\Gamma))$ preserves $\mathfrak{o} \oplus \Pi$, where Π is the purely infinite ideal, exactly when, for every $\delta > 0$, the shift- δ coefficient of its logarithmic character $\log(t^{-g}\sigma(t^g))$ vanishes on C_δ . The proof uses no exponential–logarithm correspondence and applies verbatim to the surreal field and **Oz**. Where that correspondence is available, the same condition for the operator logarithm $\text{Log } \sigma$ is equivalent; the two coefficient functionals are different.

Consequences: the stabilizer is trivial exactly in Archimedean rank one; its common fixed field is the Hahn field on the bottom convex subgroup, and for $(\mathbf{No}, \mathbf{Oz})$ it is $\mathbb{R}$; in finite rank r its derived length is at most $r - 1$, with equality for $\mathbb{Q}^r$; explicit successive-scale flows generate a faithful $\mathbb{Z} \wr \mathbb{Z}$; for every set of surreal parameters the pointwise stabilizer is nonsolvable and carries a faithful action of $\mathbb{R}^\kappa$ for every set cardinal κ . Consequently no set of parameters makes the Conway monomials, the omega-map or any valuation-representative rule definable in $(\mathbf{No}, +, \cdot, <, \mathbf{Oz})$, although the pair defines the three zero-cut projections and the floor map. A second part proves algebraic-parameter rigidity of **Oz**: embeddings into polynomial, Laurent and finite-type domains land in the coefficients, strong cancellation holds, and $\text{ML}(\mathbf{Oz}[T_1, \dots, T_n]) = \mathbf{Oz}$, while explicit commuting derivations integrate to formal flows that no polynomial family realizes.

This report merges four independently written manuscripts, prints shared results once with every source named, and keeps every source’s limitations. It is an AI-assisted draft: it has not been refereed, and nothing in it has been formalized in Lean.

Sources: manuscripts 09 (the base text), 04 and 02, pinned to repository commit `fb5c4b5`, and 08, pinned to `befe739`; their titles are in [section 1.2](#). Results already printed in the omnific Diophantine report `docs/surreal/omnific-diophantine-geometry` are cited, not reprinted. The provenance of every section is recorded in [section A](#); non-claims are collected in [section B](#). Finite checks accompany the report and check finite identities only ([section C](#)).

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1 Scope, results, and conventions

A Conway normal form carries three kinds of information: coefficients, exponents, and the monomials that represent the exponents. The omnific integers keep the purely infinite part of a normal form and an ordinary integer constant coefficient. This report asks how much of the normal-form presentation is determined by the surreal field together with this distinguished ring, and which automorphisms preserve it.

The answer is asymmetric. The coefficient field, the cut between negative and nonnegative Hahn exponents, the constant coefficient and the floor map are recoverable from the pair $(\mathbf{No}, \mathbf{Oz})$; much of this was already proved in the omnific Diophantine report (section 8). The monomial cross-section is not recoverable. Even an arbitrary *set* of named parameters leaves enough automorphisms to move a monomial to an infinite series and to defeat every definable choice of valuation representatives. The mechanism is a family of derivations that change a higher Archimedean scale by a lower one without ever moving a purely infinite term across the constant-term boundary. Part I classifies these automorphisms and derives the consequences. Part II, from a separate manuscript, shows the complementary rigidity: $\mathbf{Oz}$ admits no nonconstant *algebraic* family of embeddings, although it admits formal flows of all orders.

1.1 Principal results

Bracketed numbers name the source manuscripts (section 1.2); [merge] marks statements assembled or generalized by the merge, each with a complete proof. The Hahn conventions are fixed in section 1.5: $K = k((t^\Gamma))$ with well-ordered supports, $t^\gamma = \omega^{-\gamma}$ on the surreals, Π the series supported in $\Gamma^{<0}$, $\mathcal{A}_k = k \oplus \Pi$ and $\mathcal{R}_\circ = \circ \oplus \Pi$, so that $\mathcal{R}_\mathbb{Z} = \mathbf{Oz}$ when $k = \mathbb{R}$ and $\Gamma = \mathbf{No}$.

- (1) **The convex-support criterion** [04, 09][merge]. For every characteristic-zero k , every unital $\circ \subseteq k$, every nonzero divisible Γ (set-sized, or $\Gamma = \mathbf{No}$), and every strong k -linear 1-automorphism σ : $\sigma(\mathcal{R}_\circ) = \mathcal{R}_\circ$ if and only if the logarithmic-character coefficients $a_{\sigma, \delta}$ vanish on C_δ for every $\delta > 0$, if and only if σ commutes with the three zero-cut projections (theorem 3.3). The proof is 04's and imports no exponential-logarithm correspondence. For contracting strong derivations the analogous criterion holds with 09's coefficients c_δ (theorem 3.4); where the correspondence is available the operator form follows (corollary 3.5). The functionals $a_{\sigma, \delta}$ and c_δ differ; their vanishing conditions are equivalent (example 3.2). For single fixed-shift flows 08 proves the criterion without divisibility (theorem 4.10).
- (2) **Rank and fixed fields** [04, 08, 09]. The stabilizer $\mathcal{U}_{\mathcal{R}_\circ}(\Gamma)$ is trivial exactly when Γ is Archimedean (theorem 4.13); for set-sized Γ its common fixed field is the Hahn field on the bottom convex subgroup (theorem 4.15); and $\text{Fix}(\mathcal{U}_{\mathbf{Oz}}) = \mathbb{R}$, also for all field automorphisms of $\mathbf{No}$ preserving $\mathbf{Oz}$ (theorem 7.2). For every set A of parameters the elementary shears fixing A have common fixed field $\mathbb{R}((t^{V_A}))$, V_A the rational span of the supports (theorem 7.4).
- (3) **Group structure** [04, 08, 09]. In finite ordered rank r the stabilizer and its Lie algebra have derived length at most $r - 1$, with equality for $\mathbb{Q}^r$ (theorems 6.3 and 6.6). Two successive-scale flows generate a faithful $\mathbb{Z} \wr \mathbb{Z}$ with exact commutator leading terms (theorems 6.8 and 6.10). For every set P of surreal parameters, the pointwise stabilizer of P in $\mathcal{U}_{\mathbf{Oz}}$ carries a faithful action of the full product $\mathbb{R}^\kappa$ for every set cardinal κ and is nonsolvable (theorems 7.6 and 7.7).

- (4) **What the pair remembers** [08, 09]. Beyond the reconstruction package of the Diophantine report, the three zero-cut projections P_-, P_0, P_+ on all of $\mathbf{No}$, the floor map and truncation at a named monomial are parameter-free definable in $(\mathbf{No}, \mathbf{Oz})$, and every field automorphism preserving $\mathbf{Oz}$ commutes with them, without strongness ([theorem 8.2](#) and [corollary 5.2](#)).
- (5) **What it forgets** [04, 08, 09]. For every set P of parameters, some strong 1-automorphism preserving $\mathbf{Oz}$ fixes P pointwise and moves a Conway monomial to an infinite series; so neither the monomial class nor the omega-map is P -definable ([theorem 9.1](#)), and no P -definable rule selects a representative of each natural-valuation class ([theorem 9.4](#)).
- (6) **Surcomplex consequences** [04, 08, 09]. The criterion holds for $\mathbf{Oz}[i]$; real shears commute with conjugation; an imaginary-time shear fixing any set moves the real axis inside the leading-term kernel ([theorem 10.2](#)); a wild coefficient lift moves an ordinary real ([proposition 10.3](#)); and the common fixed field of all automorphisms preserving $\mathbf{Oz}[i]$ is $\mathbb{Q}$ ([corollary 10.4](#)).
- (7) **Algebraic-parameter rigidity** [02]. Every nonzero element of $\mathbf{Oz}$ or $\mathbf{Oz}[i]$ divides a monomial whose roots of all orders lie in the ring ([corollary 13.3](#)). Consequently injective homomorphisms into polynomial and Laurent rings over a domain land in the coefficients ([theorems 15.1](#) and [15.3](#)), polynomial cylinders over $\mathbf{Oz}$ recover $\mathbf{Oz}$ under every isomorphism ([theorem 15.4](#)), $\text{ML}(\mathbf{Oz}[\mathbf{T}]) = \mathbf{Oz}$ ([theorem 16.2](#)), and embeddings of set-sized root-covered rings into finite-type domains land in a finite field of algebraic constants ([theorem 14.4](#)). Commuting derivations of $\mathbf{Oz}$ integrate to formal flows of all orders that no polynomial family realizes, with an exact transcendence degree for their monomial orbit fields ([theorems 17.3](#) and [17.5](#)).

These statements are proved here, but proving a statement here is not a claim that nobody proved it earlier. The general theory of strong Hahn-field automorphisms and of the exponential-derivation correspondence is prior work ([section 1.3](#)); each source names its most distinctive contributions, and no exhaustive priority claim is made ([section B](#)).

1.2 Provenance of this report

The report is built from four manuscripts written independently and placed together in commit `f4c9504` (batch 26, items 02, 04, 08 and 09; they keep these numbers here, as their shipped file prefixes do). Manuscripts 02, 04 and 09 pin repository commit `fb5c4b53` (12 commits before the placement); manuscript 08 pins `befe739b` (10 commits before). Both pins contain the batch-24 placement `be06fc8` but precede the batch-24 and batch-25 writes, so no source saw the reconstruction package now in the omnific Diophantine report (its section `odg:def:sec:reconstruction`, placed after both pins).

Source	Manuscript	Principal contribution to this report
09	<i>Omnific-Preserving Automorphisms: convex-scale stabilizers, definable constants, and nondefinable monomials</i>	Base text and structure; the criterion for any characteristic-zero k and unital $\mathfrak{o}$; the derivation form with c_δ ; the Archimedean dichotomy; the strong factorization; the finite-rank derived-length bound and its sharpness for the abstract group; set-parameter nondefinability and the absence of any definable representative rule; $\mathbb{R}^\kappa$ actions and nonsolvable stabilizers; projections and truncation.
04	<i>Automorphisms Preserving the Omnific Integers: a convex-support criterion, fixed fields, and nondefinability of Conway monomials</i>	The import-free proof of the criterion through the logarithmic character (the proof of record here); exact displacement and fixed fields; the bottom subgroup and the relative Hahn hull; the common-positive-shift exponential–logarithm correspondence; the explicit commutator formula; the one-term class and the RV sort; the real axis inside the leading-term kernel.
08	<i>What the Omnific Integer Part Remembers: coefficient reconstruction, convex-scale automorphisms, nondefinable monomials, and an explicit wreath product</i>	The divisibility-free threshold for single fixed-shift flows; phase shears; the faithful $\mathbb{Z} \wr \mathbb{Z}$ and exact commutator leading terms; the floor map and the order formula on $\mathbf{Oz}$; the wild coefficient lift and the Gaussian fixed field $\mathbb{Q}$.
02	<i>Algebraic-Parameter Rigidity of the Omnific Integers: strong cancellation, finite-type embeddings, and nonalgebraizable formal flows</i>	All of part II .

Manuscripts 04, 08 and 09 prove one classification at three generalities ([section 3](#)) and the same consequences: rank triviality, fixed field $\mathbb{R}$, set-parameter nondefinability and a nonnilpotent stabilizer. They are printed once. 09 is the base because its hypotheses on $(k, \mathfrak{o})$ are the weakest and its group-theoretic results the widest; 04’s proof is used for the criterion itself because it needs no imported correspondence, and 08’s single-flow version is kept as a marked case because it needs no divisibility. Their reconstruction and definability sections re-derive results now in the Diophantine report; those are cited, and only the remaining extras are printed ([section 8](#)). Manuscript 02 has a different subject, algebraic families rather than automorphisms preserving a support condition, and forms [part II](#) with the sub-prefix `opa:par::`; its relation to [part I](#) is recorded in [section 19](#). None of the four contradicts another or any report in the collection. [Section A](#) records, section by section, what each source proved, where the merge had to choose, and every symbol renamed from a source.

1.3 Imported results

Conway normal forms, real closedness of $\mathbf{No}$ and the omnific integers are classical [[4](#), [5](#)]; factorization theory of $\mathbf{Oz}$ is due to L’Innocente and Mantova [[9](#)]. The monomial/character/kernel

decomposition of valued Hahn automorphisms is due to Kuhlmann and Serra [8], and its surreal form, with the class-size conventions used here, to Kaplan, Krapp and Serra [7]. The collection’s surcomplex automorphism report (`docs/surcomplex/surcomplex-field-automorphisms`) already constructs the fixed-shift derivations and flows used below, in greater generality (its `saut:thm:shiftflow`), and the four-layer decomposition of valued automorphisms (`saut:thm:decomp`).

The correspondence between contracting strong derivations and strong 1-automorphisms is due to Bagayoko, Krapp, Kuhlmann, Panazzolo and Serra.

Cited theorem 1.1 ([1, Theorem 3.13]; [7, Fact 4.2]). *Let k have characteristic zero and Γ be an ordered abelian group. If ∂ is a contracting strongly k -linear derivation of $k((t^\Gamma))$, then*

$$\text{Exp}(\partial)(f) = \sum_{n \geq 0} \frac{\partial^n(f)}{n!}$$

is a strongly k -linear 1-automorphism, and $\partial \mapsto \text{Exp}(\partial)$ is a bijection between contracting strongly k -linear derivations and strongly k -linear 1-automorphisms, with inverse $\text{Log}(\sigma) = \sum_{n \geq 1} (-1)^{n+1} (\sigma - \text{id})^n / n$. Composition corresponds to the Baker–Campbell–Hausdorff operation.

Remark 1.2 (What is proved where). [1] proves the correspondence for set-sized Γ . Source 09 attributes the surreal case to [7, Fact 4.2]; the placement dossier, reading an HTML rendering, doubted that this Fact states the bijection. The merge re-read the PDF of [7] (version 3): Fact 4.2 does state the bijection, cites [1, Theorem 3.13], and adds that the correspondence “was established in the case that G is a set, but it also holds for G a proper class”, without a separate proof. So 09’s citation is accurate, and for $\Gamma = \mathbf{No}$ the operator-logarithm statements rest on that assertion. The criterion itself does not: its proof (theorem 3.3) and 04’s correspondence on the common-positive-shift class (theorem 4.8) are complete and elementwise, and the explicit constructions on $\mathbf{No}$ use one set-sized well-ordered set of shifts (proposition 4.4). Every statement below that uses cited theorem 1.1 says so.

1.4 Class size and foundations

Unless $\Gamma = \mathbf{No}$ is explicitly under discussion, exponent groups are sets, so fields and automorphism groups are ordinary sets. For $\Gamma = \mathbf{No}$ we work in NBG with global choice [04, 08, 09]; every individual Hahn support and every summation family is a set. An automorphism of $\mathbf{No}$ is an individual class map; no set or NBG class whose members are all such proper-class graphs is formed. A statement about $\mathcal{U}_{\mathbf{Oz}}$ is a statement about individual maps, their compositions, or a set-indexed action encoded by one class relation; this is the convention of [7]. A common-fixed-field statement is pointwise; nonsolvability means that no ordinary finite derived length annihilates all iterated commutators; valuation representatives are treated through an equivalence relation on nonzero elements, not through proper-class cosets [09]. A quotient notation G/H in a full-class application abbreviates an additive map on G vanishing on H [08]. Bases are chosen only for set-sized vector spaces (lemma 7.3).

1.5 Notation and conventions

Part I uses the small- t Hahn convention of sources 04 and 09. Part II keeps its source’s large-monomial convention, which is the convention of the other omnific reports; see section 12.1. For actual surreals

$$t^\gamma = \omega^{-\gamma}. \tag{1.1}$$

Thus negative t -exponents are purely infinite terms and positive ones are infinitesimal. The tempting false reading is the opposite: in this Part, Π is supported at *negative* t -exponents, and a “positive shift” $\delta > 0$ moves a term towards the infinitesimal side. Source 08 wrote in the ω -convention with decreasing supports; its statements have been translated, and its derivation $D_{\varphi,\delta}$ with $D_{\varphi,\delta}(\omega^g) = \varphi(g)\omega^{g-\delta}$ is $D_{\delta,\theta}$ below with $\theta(\gamma) = \varphi(-\gamma) = -\varphi(\gamma)$.

Notation	Meaning in part I , and the sources’ symbols
$K = k((t^\Gamma))$	Hahn field, k of characteristic zero; $v(f) = \min \text{supp } f$, $v(0) = \infty$; $\text{lc}(f)$ the coefficient at $v(f)$.
Π	Series supported in $\Gamma^{<0}$, zero included (the purely infinite ideal). Sources: $\mathcal{J}$ (08, 09), $\mathcal{I}$ (04).
$\mathcal{A}_k = k \oplus \Pi$	Series supported in $\Gamma^{\leq 0}$, the boundary ring. Sources: B (09), $\mathcal{A}$ (04), $\mathcal{B}_k$ (08); the Diophantine report’s $\mathcal{A}_{\mathbb{k}}(\Gamma)$.
$\mathcal{R}_{\mathfrak{o}} = \mathfrak{o} \oplus \Pi$	The integer-part ring for a unital subring $\mathfrak{o} \subseteq k$; $\mathcal{R}_{\mathbb{Z}} = \mathbf{Oz}$ for $(\mathbb{R}, \mathbf{No})$ and $\mathcal{R}_{\mathbb{Z}[i]} = \mathbf{Oz}[i]$ for $(\mathbb{C}, \mathbf{No})$. Sources: A_D (09), O_D (04), A_k (08); the Diophantine report’s $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$. The letter D of the sources is freed for derivations.
$\mathcal{O}, \mathfrak{m}$	Series supported in $\Gamma^{\geq 0}$ and $\Gamma^{>0}$: the valuation ring and its maximal ideal.
P_-, P_0, P_+	Projections onto Π , k and $\mathfrak{m}$ in $K = \Pi \oplus k \oplus \mathfrak{m}$; P_0 restricted to $\mathcal{A}_k$ is the retraction ct .
C_δ	Principal convex subgroup $\{\gamma : \gamma \leq n\delta \text{ for some } n \geq 1\}$ of $\delta > 0$. Sources: C_δ (04, 09), H_δ (08), $H(a)$ (02); the transcendence report’s $C(a)$.
$\delta \ll \epsilon$	$n\delta < \epsilon$ for every ordinary $n \geq 1$.
H_g	$\{h : n h < g \text{ for all } n \geq 1\}$ for $g \neq 0$ [04]; the same convex subgroup as the quotient report’s H_a with $a = g $.
$a_{\sigma,\delta}$	Logarithmic-character coefficient $[t^\delta] \log(t^{-g}\sigma(t^g))$ of a 1-automorphism [04].
c_δ	Operator coefficient $[t^{\gamma+\delta}] \partial(t^\gamma)$ of a derivation ∂ [09]; for $\partial = \text{Log } \sigma$ it differs from $a_{\sigma,\delta}$ (example 3.2).
$D_{\delta,\theta}$	The derivation $t^\gamma \mapsto \theta(\gamma)t^{\gamma+\delta}$ for additive $\theta : \Gamma \rightarrow k$ and $\delta > 0$. Sources: $D_{\delta,\theta}$ (09), $D_{\delta,a}$ (04), $D_{\varphi,\delta}$ (08, sign as above).
$\mathcal{U}_{\mathcal{R}_{\mathfrak{o}}}(\Gamma), \mathcal{U}_{\mathbf{Oz}}$	Strong k -linear 1-automorphisms preserving $\mathcal{R}_{\mathfrak{o}}$ (independent of $\mathfrak{o}$ by theorem 3.3). Sources: $\mathcal{U}_{A_D}$ (09), $\mathcal{U}_D(G)$ (04).
$q_\beta(\gamma)$	The coefficient of ω^β in the Conway normal form of the <i>exponent</i> $\gamma \in \mathbf{No}$. Sources: q_η (09), $[\omega^\beta]g$ (04), c_a (08), χ_b (02), ℓ_η (quotient report).
$\exp(u), \text{Exp}, \text{Log}$	The Hahn series $\sum u^n/n!$ for $v(u) > 0$ (E in 08); operator exponential and logarithm.

Two further collisions are resolved in the new material. Source 09’s parameter monomial $H = \omega^\eta$ and its family H_j are written $\eta = \omega^\beta$ and η_j here, leaving H_g to 04’s convex subgroup; and 08’s automorphisms A_1, B_1 are written in 04’s normalization A_s, B_u , with $A_1^{(08)} = A_{-1}$

and $B_1^{(08)} = B_{-1}$ (section 6). A 1-automorphism is a field automorphism that preserves the valuation and the leading coefficient of every nonzero series. *Strong* means that every Hahn-summable family is sent to a summable family with the same sum; a strong automorphism is one whose inverse is also strong (04's convention); *k-linear* includes fixing k pointwise.

Part I

Automorphisms preserving the omnific integers

2 Hahn support, the boundary ring, and strong maps

2.1 Summability

An element of $K = k((t^\Gamma))$ is a formal sum $f = \sum_{\gamma \in S} a_\gamma t^\gamma$ with $S \subseteq \Gamma$ well ordered. A set-indexed family $(f_i)_{i \in I}$ is *Hahn summable* if the union of its supports is well ordered and each exponent lies in only finitely many of the supports; its sum is formed coefficientwise with finite sums in k . No convergence in the fine topology of the surreals is involved; a strong sum is a support-controlled algebraic construction [04, 08, 09].

We use the Neumann support principle in its word form: if S is well ordered and $T \subseteq \Gamma^{>0}$ is well ordered, then $S + \langle T \rangle_{\mathbb{N}}$ is well ordered, and each exponent receives only finitely many contributions from a choice of an element of S and a finite word in T , counted across all word lengths [1]. The one-shift case is proved directly.

Lemma 2.1 (One-shift summation [04, 08]). *If $T \subseteq \Gamma$ is well ordered and $\delta > 0$, then $T + \mathbb{N}\delta$ is well ordered, and each $\gamma \in \Gamma$ has only finitely many representations $\gamma = g + n\delta$ with $g \in T$ and $n \in \mathbb{N}$.*

Proof. An infinite strictly decreasing sequence $g_j + n_j\delta$ has an infinite subsequence on which the n_j are nondecreasing; on it the g_j strictly decrease, contradicting well-ordering of T . Infinitely many representations of one γ would have unbounded, hence eventually increasing, values of n , and the corresponding $g = \gamma - n\delta$ would strictly decrease in T . $\square$

For $h \in \mathfrak{m}$ the series $\exp(h) = \sum_{n \geq 0} h^n/n!$ and $\log(1+h) = \sum_{n \geq 1} (-1)^{n+1} h^n/n$ are Hahn summable, and they are inverse group isomorphisms between $(\mathfrak{m}, +)$ and $(1 + \mathfrak{m}, \cdot)$. This local exponential is the only exponential used in the criterion; neither $\exp$ nor Exp asserts anything about the Gonshor exponential of $\mathbf{No}$.

Lemma 2.2 (Coefficient polynomial [04]). *For $h \in \mathfrak{m}$ and $\delta > 0$ the expression $P_{h,\delta}(X) = \sum_{n \geq 1} \frac{X^n}{n!} [t^\delta] h^n$ is a polynomial over k . For every $q \in \mathbb{Q}$ one has $P_{h,\delta}(q) = [t^\delta] \exp(qh)$, and its coefficient of X is $[t^\delta] h$.*

Proof. The support principle gives only finitely many contributions to the exponent δ across all powers h^n , not merely within each power. So only finitely many displayed terms are nonzero. The rest is coefficientwise. $\square$

2.2 The rings and the zero cut

The additive decompositions

$$K = \Pi \oplus k \oplus \mathfrak{m}, \quad \mathcal{A}_k = k \oplus \Pi, \quad \mathcal{R}_0 = \mathfrak{o} \oplus \Pi, \quad \mathcal{O} = k \oplus \mathfrak{m} \quad (2.1)$$

define the projections P_-, P_0, P_+ [04, 09]. Membership in Π means that *all* exponents are negative, which is stronger than $v(f) < 0$: the series $t^{-1} + t$ has negative valuation and is not in Π [04]. Π is a nonunital ideal of $\mathcal{A}_k$ and of $\mathcal{R}_o$; $\mathcal{A}_k$ is not a valuation ring; P_0 is a ring retraction $\mathcal{A}_k \rightarrow k$ restricting to $\mathcal{R}_o \rightarrow o$, but not a ring map on K : $P_0(t^{-1})P_0(t) = 0 \neq 1 = P_0(1)$. For $k = \mathbb{R}$, $o = \mathbb{Z}$ the ring $\mathcal{R}_{\mathbb{Z}}$ is an integer part of K ; for $\Gamma = \mathbf{No}$ it is $\mathbf{Oz}$ and its floor function is the Diophantine report's `odg:thm:floor`.

The units $\mathcal{A}_k^\times = k^\times$, and more generally the fact that a product of two elements of $\mathcal{A}_k$ lying in $k^\times$ has constant factors, are the Diophantine report's constant-product lemma (`odg:lem:constantproduct`, `odg:def:lem:units`); sources 04, 08 and 09 reprove it, and it is not reprinted.

Lemma 2.3 (The divisible part [04, 08]). *If $\bigcap_{n \geq 1} n o = 0$ (for example $o = \mathbb{Z}$ or $\mathbb{Z}[i]$), then $\Pi = \bigcap_{n \geq 1} n \mathcal{R}_o$, the largest divisible subgroup of $\mathcal{R}_o$; it is preserved by every ring automorphism of $\mathcal{R}_o$. This holds for every ordered abelian Γ , divisible or not.*

Proof. A series in Π is n times a series with the same support. Conversely the constant coefficient of an element of the intersection lies in $\bigcap_n n o = 0$. Every divisible subgroup lies in the intersection. $\square$

For $\mathbf{Oz}$ this is the identity `odg:eq:divkernel` of the Diophantine report; it is printed because the threshold theorem uses it for non-divisible Γ .

Definition 2.4 (Principal convex scale). For $\delta > 0$ let $C_\delta = \{\gamma \in \Gamma : |\gamma| \leq n\delta \text{ for some } n \geq 1\}$, and write $\delta \ll \epsilon$ if $n\delta < \epsilon$ for all $n \geq 1$. Then $\gamma \notin C_\delta$ if and only if $\delta \ll |\gamma|$; $C_\delta \subsetneq C_\epsilon$ if and only if $\delta \ll \epsilon$; and C_δ is divisible when Γ is. For $g \neq 0$ the set $H_g = \{h : n|h| < |g| \text{ for all } n \geq 1\}$ is a convex subgroup, and a finite sum of positive elements of H_g is still smaller than $|g|$ [04].

3 The convex-support criterion

3.1 Two coefficient functionals

Let σ be a strong k -linear 1-automorphism. Since $t^{-g}\sigma(t^g)$ has leading term 1, its local logarithm is defined [04]:

$$h_\sigma(g) = \log(t^{-g}\sigma(t^g)) \in \mathfrak{m}, \quad a_{\sigma,\delta}(g) = [t^\delta]h_\sigma(g) \quad (\delta > 0), \quad (3.1)$$

so that $\sigma(t^g) = t^g \exp(h_\sigma(g))$. For a contracting strong k -derivation ∂ (that is, $v(\partial f) > v(f)$ for $f \neq 0$) put [09]

$$c_\delta(\gamma) = [t^{\gamma+\delta}]\partial(t^\gamma) \quad (\delta > 0), \quad \text{so} \quad \partial(t^\gamma) = t^\gamma \sum_{\delta > 0} c_\delta(\gamma) t^\delta. \quad (3.2)$$

For fixed γ the nonzero terms have well-ordered set support; no assumption is made that all shifts together form one well-ordered set.

Lemma 3.1 (Additivity [04, 09]). *The maps $h_\sigma : \Gamma \rightarrow (\mathfrak{m}, +)$, $a_{\sigma,\delta}$ and c_δ are additive, hence $\mathbb{Q}$ -linear when Γ is divisible. If Γ is divisible and an additive $\theta : \Gamma \rightarrow k$ vanishes on $(-\delta, 0)$, it vanishes on C_δ .*

Proof. The quotient at $g + g'$ is the product of the quotients at g and g' , and log takes products to sums. For c_δ the product rule gives $t^{-\gamma-\eta}\partial(t^{\gamma+\eta}) = t^{-\gamma}\partial(t^\gamma) + t^{-\eta}\partial(t^\eta)$. For the last claim, given $0 \neq h \in C_\delta$ choose N with $|h|/N < \delta$; one of $\pm h/N$ lies in $(-\delta, 0)$, and additivity gives $\theta(h) = 0$. $\square$

Example 3.2 (The functionals differ [merge]). Let $\Gamma = \mathbf{No}$, $k = \mathbb{R}$, and $\partial = D_{1,q_1} + D_{\omega,q_2}$ (notation of [section 1.5](#); the t -exponent γ has $q_1(\gamma)$ and $q_2(\gamma)$ as its coefficients at ω and ω^2). Take $\sigma = \text{Exp}(\partial)$, an element of $\mathcal{U}_{\mathbf{Oz}}$ by [proposition 4.4](#), and $g = \omega^2$. Then $\partial(t^g) = t^{g+\omega}$ and $\partial^2(t^g) = t^{g+\omega+1} + t^{g+2\omega}$, and the only word in the shifts $\{1, \omega\}$ reaching the exponent $\omega + 1$ applies the shift ω first; $\log(1 + u) = u - u^2/2 + \dots$ adds nothing at $\omega + 1$, since u^2 starts at 2ω . Hence

$$t^{-g}\sigma(t^g) = 1 + t^\omega + \frac{1}{2}t^{\omega+1} + (\text{terms at exponents } > \omega + 1), \quad a_{\sigma, \omega+1}(\omega^2) = \frac{1}{2},$$

whereas $c_{\omega+1}(\omega^2) = 0$: the operator coefficients of ∂ at g are $c_\omega(g) = 1$ and zero at every other shift. So $a_{\sigma, \delta} \neq c_\delta$ in general; 04's condition concerns the logarithm of the monomial image, 09's the logarithm of the operator. Both vanishing conditions hold here, and by [theorem 3.3](#) and [corollary 3.5](#) each is equivalent to preservation of $\mathbf{Oz}$, hence to the other. The placement dossier checked the example by exact computation.

3.2 The criterion

The following statement combines 09's generality with 04's proof. 04 proved it for $k = \mathbb{R}, \mathbb{C}$ and $\mathfrak{o} = \mathbb{Z}, \mathbb{Z}[i]$, passing through [lemma 2.3](#); 09 proved it for every characteristic-zero k and unital $\mathfrak{o}$, but through the operator logarithm ([cited theorem 1.1](#)). The first step below is 09's and removes the need for $\bigcap n\mathfrak{o} = 0$; the rest is 04's.

Theorem 3.3 (Convex-support criterion [04, 09][merge]). *Let k have characteristic zero, let $\mathfrak{o} \subseteq k$ be a unital subring, let $\Gamma \neq 0$ be divisible (a set, or $\Gamma = \mathbf{No}$), and let σ be a strong k -linear 1-automorphism of $K = k((t^\Gamma))$. The following are equivalent:*

- (i) $\sigma(\mathcal{R}_{\mathfrak{o}}) = \mathcal{R}_{\mathfrak{o}}$;
- (ii) $\sigma(\mathcal{A}_k) \subseteq \mathcal{A}_k$;
- (iii) $\sigma(\Pi) \subseteq \Pi$;
- (iv) $a_{\sigma, \delta}(C_\delta) = 0$ for every $\delta > 0$;
- (v) for every $g \neq 0$, every exponent $\delta \in \text{supp } h_\sigma(g)$ satisfies $\delta \ll |g|$;
- (vi) σ commutes with P_- , P_0 and P_+ .

When they hold, $\sigma(\Pi) = \Pi$, $\sigma(\mathcal{A}_k) = \mathcal{A}_k$, $\sigma(\mathfrak{m}) = \mathfrak{m}$, the constant coefficient of every series is preserved, and every purely infinite monomial stays purely infinite, though not necessarily a monomial. In particular the group $\mathcal{U}_{\mathcal{R}_{\mathfrak{o}}}(\Gamma)$ does not depend on $\mathfrak{o}$.

Proof. (i) $\Rightarrow$ (ii) [09]. For $\gamma < 0$, $t^\gamma \in \mathcal{R}_{\mathfrak{o}}$, so $\sigma(t^\gamma) \in \mathcal{R}_{\mathfrak{o}} \subseteq \mathcal{A}_k$. An element of $\mathcal{A}_k$ is $c + \sum_{\gamma < 0} a_\gamma t^\gamma$ with $c \in k$; strong k -linearity writes its image as $c + \sum a_\gamma \sigma(t^\gamma)$, a summable family of series supported in $\Gamma^{\leq 0}$, whose sum is supported there.

(ii) $\Rightarrow$ (iv) [04]. Fix $\delta > 0$ and $0 \neq g \in C_\delta$; put $x = -|g| < 0$, so $h_\sigma(x) = \pm h_\sigma(g)$, and choose $N \geq 1$ with $|g| \leq N\delta$. By [lemma 2.2](#), $P(X) = [t^\delta] \exp(Xh_\sigma(x))$ is a polynomial. For rational $0 < q < 1/N$, divisibility gives $qx \in \Gamma^{< 0}$ and additivity gives $\sigma(t^{qx}) = t^{qx} \exp(qh_\sigma(x))$, whose coefficient at $qx + \delta$ is $P(q)$. Since $q|g| < \delta$, this exponent is strictly positive, while $\sigma(t^{qx}) \in \mathcal{A}_k$ has no positive exponents. So P has infinitely many zeros and vanishes; its linear coefficient is $a_{\sigma, \delta}(x)$, so $a_{\sigma, \delta}(g) = 0$. This tests every coefficient of the logarithm; testing only the first nonzero one would not suffice.

(iv) $\Leftrightarrow$ (v) [04]. $g \notin C_\delta$ means $\delta \ll |g|$; consider each nonzero coefficient.

(v) $\Rightarrow$ (iii) [04]. For $g < 0$, $\text{supp } h_\sigma(g) \subseteq H_g^{>0}$, so every exponent of $\exp(h_\sigma(g))$ is 0 or a finite sum of elements of $H_g^{>0}$, hence smaller than $|g|$; thus $\sigma(t^g) = t^g \exp(h_\sigma(g)) \in \Pi$. For $f \in \Pi$ write $\sigma(f) = \sum c_g \sigma(t^g)$ by strongness; all supports are negative.

(iii) $\Rightarrow$ (vi) [04]. A 1-automorphism maps $\mathfrak{m}$ into $\mathfrak{m}$, and σ fixes k ; so each summand of (2.1) is mapped into itself and σ commutes with the projections.

(vi) $\Rightarrow$ (i) [04]. If $\sigma(y) \in \Pi$ with $y = y_- + c + y_+$, then $0 = P_0 \sigma(y) = c$ and $0 = P_+ \sigma(y) = \sigma(y_+)$, so $y \in \Pi$. With (iii) this gives $\sigma(\Pi) = \Pi$, and since σ fixes $\mathfrak{o} \subseteq k$, $\sigma(\mathcal{R}_\mathfrak{o}) = \mathcal{R}_\mathfrak{o}$. The same argument gives $\sigma(\mathcal{A}_k) = \mathcal{A}_k$ and $\sigma(\mathfrak{m}) = \mathfrak{m}$, and (vi) preserves constant coefficients. Condition (ii) does not mention $\mathfrak{o}$, which proves the last assertion. $\square$

Only (ii) $\Rightarrow$ (iv) uses divisibility; it divides an *exponent*, while characteristic zero separately permits rational scalars [09]. The argument is elementwise with set-sized supports, so it applies verbatim to $\Gamma = \mathbf{No}$, $k = \mathbb{R}$, $\mathfrak{o} = \mathbb{Z}$, that is to $\mathbf{Oz}$, and to $\mathbf{Oz}[i]$ with $(k, \mathfrak{o}) = (\mathbb{C}, \mathbb{Z}[i])$ [04]. The theorem classifies *existing* strong maps: an arbitrary additive $h : \Gamma \rightarrow \mathfrak{m}$ need not define a strong surjective field map by $t^g \mapsto t^g \exp h(g)$ [04] (question 11.2).

Theorem 3.4 (Boundary-preserving derivations [09]). *Let $\Gamma \neq 0$ be divisible and let ∂ be a contracting strong k -derivation of K . Then $\partial(\mathcal{A}_k) \subseteq \mathcal{A}_k$ if and only if $c_\delta(C_\delta) = 0$ for every $\delta > 0$. In that case $\partial(\mathcal{A}_k) \subseteq \Pi$, $\partial(\Pi) \subseteq \Pi$, and ∂ commutes with P_- , P_0 , P_+ .*

Proof. If $\partial(\mathcal{A}_k) \subseteq \mathcal{A}_k$ and $-\delta < \gamma < 0$, then $t^\gamma \in \mathcal{A}_k$ and $\gamma + \delta > 0$, so $c_\delta(\gamma) = 0$; lemma 3.1 extends this to C_δ . Conversely, if $\gamma < 0$ and $c_\delta(\gamma) \neq 0$, then $\gamma \notin C_\delta$, so $-\gamma > n\delta$ for all n and in particular $\gamma + \delta < 0$. Every term of $\partial(t^\gamma)$ is negative; constants are killed; strongness sums over any negative support. Positive exponents stay positive under positive shifts, which gives the projection identities. $\square$

The condition excludes a boundary term as well as a crossing term: testing $-\delta/2$ gives $c_\delta(-\delta) = 0$. This matters when constant coefficients are restricted to $\mathfrak{o}$ rather than k [09]. Source 04 proves the same criterion for derivations with common positive shifts (theorem 4.7), testing t^{-u} for $0 < u \leq \delta$.

Corollary 3.5 (Operator form [09]). *Let σ be a strong k -linear 1-automorphism of the form $\sigma = \text{Exp}(\partial)$ for a contracting strong k -derivation ∂ with $\text{Log}(\sigma) = \partial$. Then the conditions of theorem 3.3 are equivalent to $\partial(\mathcal{A}_k) \subseteq \mathcal{A}_k$ and to $c_\delta(C_\delta) = 0$ for all $\delta > 0$. For set-sized Γ every strong k -linear 1-automorphism has this form, so*

$$\mathcal{U}_{\mathcal{R}_\mathfrak{o}}(\Gamma) = \text{Exp}(\mathfrak{L}_A), \quad \mathfrak{L}_A = \{\partial \text{ contracting strong } k\text{-derivation} : c_\delta(C_\delta) = 0 \text{ for all } \delta > 0\}.$$

For $\Gamma = \mathbf{No}$ the same equality holds if the correspondence of cited theorem 1.1 is granted for proper-class Γ , as [7, Fact 4.2] asserts (remark 1.2); on the common-positive-shift class it is proved in theorem 4.8.

Proof. If the conditions hold, $\sigma(\mathcal{A}_k) = \mathcal{A}_k$; then each $(\sigma - \text{id})^n$ maps $\mathcal{A}_k$ into $\mathcal{A}_k$, and $\text{Log}(\sigma)$ is a strongly summable family of series supported in $\Gamma^{\leq 0}$, so $\partial(\mathcal{A}_k) \subseteq \mathcal{A}_k$. This step is taken in the k -algebra $\mathcal{A}_k$, not in $\mathcal{R}_\mathfrak{o}$, so no division by integers inside $\mathfrak{o}$ is needed. Theorem 3.4 converts this into the coefficient condition. Conversely, under the coefficient condition every iterate of ∂ maps $\mathcal{A}_k$ into Π and kills k , so $\text{Exp}(\pm\partial)$ preserve $\mathcal{R}_\mathfrak{o}$. The last assertions are cited theorem 1.1. $\square$

The two routes are genuinely different. 04's functional is read off the monomial images and needs no correspondence; 09's is read off the generator and gives the Lie-theoretic structure of section 6. Neither condition is stated in terms of the other: each is equivalent to preservation of $\mathcal{R}_\mathfrak{o}$ (example 3.2).

Remark 3.6 (What has and has not been classified [04, 09]). The classification is inside the *strong, coefficient-fixing, leading-term-preserving* automorphisms. It does not assert that every abstract field automorphism preserving $\mathcal{R}_o$ is strong; for the real pair, fixing of $\mathbb{R}$ is automatic (corollary 5.2), strongness is not known to be. The coefficient functionals come from an existing strong map; arbitrary functions satisfying the vanishing condition cannot be assembled into an operator without checking summability and invertibility.

4 Shears, flows, and the rank dichotomy

4.1 Homogeneous shears

Let $\theta : \Gamma \rightarrow (k, +)$ be additive and $\delta > 0$. The formula

$$D_{\delta, \theta} \left(\sum_{\gamma} a_{\gamma} t^{\gamma} \right) = \sum_{\gamma} a_{\gamma} \theta(\gamma) t^{\gamma + \delta} \quad (4.1)$$

defines a contracting strong derivation: translation keeps supports well ordered and families summable, and additivity of θ gives the product rule on monomials, hence on all series. This is the fixed-shift derivation of the surcomplex automorphism report, `saut:thm:shiftflow`, which constructs its exponential for every additive θ and every $\delta > 0$; its n th iterate has coefficient $\prod_{j < n} \theta(\gamma + j\delta)$. Sources 04, 08 and 09 re-derive it; only 04 credited it. The integer-part condition is the additional restriction $\theta(C_{\delta}) = 0$, not merely $\theta(\delta) = 0$.

Theorem 4.1 (Homogeneous shears [04, 09]; after `saut:thm:shiftflow`). *Suppose $\theta(C_{\delta}) = 0$. Then $D_{\delta, \theta}^n(t^{\gamma}) = \theta(\gamma)^n t^{\gamma + n\delta}$ and*

$$\sigma_{\delta, \theta}(t^{\gamma}) := \text{Exp}(D_{\delta, \theta})(t^{\gamma}) = t^{\gamma} \exp(\theta(\gamma)t^{\delta}). \quad (4.2)$$

The strong extension $\sigma_{\delta, \theta}$ lies in $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$ for every θ , with inverse $\sigma_{\delta, -\theta}$, and $\sigma_{\delta, \theta}\sigma_{\delta, \psi} = \sigma_{\delta, \theta + \psi}$ when θ, ψ both vanish on C_{δ} . With $\Phi_s = \sigma_{\delta, s\theta}$ ($s \in k$), $s \mapsto \Phi_s$ is a homomorphism, injective if $\theta \neq 0$. Conversely $\text{Exp}(D_{\delta, \theta})$ preserves $\mathcal{R}_o$ only if $\theta(C_{\delta}) = 0$.

Proof. $\theta(\delta) = 0$ gives $\theta(\gamma + n\delta) = \theta(\gamma)$ and the iterate formula. For a series supported on S , all terms of the image lie in $S + \mathbb{N}\delta$; lemma 2.1 gives well-ordering and finite contributions across all n , for single inputs and for summable families. On monomials the prescription is multiplicative by additivity of θ and the exponential law; strongness extends this to all series. The map fixes t^{δ} and every coefficient of $\exp(\cdot)$, so applying it termwise gives $\sigma_{\delta, \theta}\sigma_{\delta, \psi}(t^{\gamma}) = t^{\gamma} \exp((\theta + \psi)(\gamma)t^{\delta})$, which gives the group law and the inverse. Every correction has larger exponent, so leading terms are fixed. Here $h_{\sigma}(\gamma) = \theta(\gamma)t^{\delta}$, whose only coefficient $a_{\sigma, \delta} = \theta$ annihilates C_{δ} ; apply theorem 3.3. The coefficient at $\gamma + \delta$ of $\Phi_s(t^{\gamma})$ is $s\theta(\gamma)$, which gives injectivity. The converse is the same criterion, since $a_{\sigma, \delta} = \theta$ is the only functional. $\square$

The summability above is proved directly; no appeal to cited theorem 1.1 is made. For $\gamma < 0$ with $\theta(\gamma) \neq 0$, every $\gamma + n\delta$ stays negative, because $-\gamma > n\delta$ for all ordinary n : the whole exponential, not only a Taylor polynomial, stays on the correct side of the boundary [09].

Theorem 4.2 (Exact displacement and fixed field [04, 08, 09]). *Let $\theta(C_{\delta}) = 0$, $s \neq 0$, and $f = \sum c_{\gamma} t^{\gamma}$. If $T_{\theta}(f) = \{\gamma \in \text{supp } f : \theta(\gamma) \neq 0\}$ is empty, $\Phi_s(f) = f$. Otherwise let $\gamma_0 = \min T_{\theta}(f)$; then*

$$v(\Phi_s(f) - f) = \gamma_0 + \delta, \quad [t^{\gamma_0 + \delta}](\Phi_s(f) - f) = s\theta(\gamma_0)c_{\gamma_0}.$$

In particular $\text{Fix}(\Phi_s) = k((t^{\ker \theta}))$.

Proof. Only exponents with $\theta(\gamma) \neq 0$ and powers $n \geq 1$ contribute to the difference, at $\gamma + n\delta \geq \gamma_0 + \delta$, with equality only for $\gamma = \gamma_0$, $n = 1$. $\ker \theta$ is a divisible subgroup, so the Hahn field on it is a subfield. $\square$

This first-displacement computation appears three times in the sources: as 04's exact displacement, as 08's leading term of $T(x) - x$, and inside 09's fixed-field proof. It is printed once.

Example 4.3 (A nonmonomial image [04, 09]). Let $\Gamma = \mathbf{No}$, $\delta = 1$, $\theta = q_1$, the coefficient of ω in an exponent. Every member of C_1 is finite, so $q_1(C_1) = 0$. The flow fixes $\mathbb{R}$ and $\omega = t^{-1}$, and

$$\Phi_s(\omega^\omega) = \omega^\omega \exp(-s\omega^{-1}) = \sum_{n \geq 0} \frac{(-s)^n}{n!} \omega^{\omega-n}.$$

For $s \neq 0$ this is not a monomial; every exponent $\omega - n$ is positive, so it is an omnific integer. The sum is a Hahn sum with strictly decreasing exponents, not a limit in the fine topology.

4.2 Families of shifts and a proved correspondence

Proposition 4.4 (Uniformly supported shifts [09]). *Let $S \subseteq \Gamma^{>0}$ be a set-sized well-ordered set and $(\theta_s)_{s \in S}$ additive maps $\Gamma \rightarrow k$. Then $D(f) = \sum_{\gamma} \sum_{s \in S} a_{\gamma} \theta_s(\gamma) t^{\gamma+s}$ is a contracting strong derivation, the terms of $\text{Exp}(D)(f)$ are supported in $\text{supp } f + \langle S \rangle_{\mathbb{N}}$ and form a summable family, and if $\theta_s(C_s) = 0$ for all s then $\text{Exp}(D) \in \mathcal{U}_{\mathcal{R}_o}(\Gamma)$.*

Proof. The support principle applies to $\text{supp } f + S$ and to all finite words in S ; every coefficient of D , its iterates and products is a finite sum. Additivity of each θ_s gives the product rule; replacing one input support by the union of the supports of a summable input family gives strongness; each output exponent exceeds its input exponent. Preservation of $\mathcal{R}_o$ follows from the no-crossing argument of [theorem 3.4](#) applied at each step of a word in S , which shows that $\text{Exp}(\pm D)$ preserve Π and fix k . $\square$

All explicit constructions on $\mathbf{No}$ below use one set-sized well-ordered set of shifts, so they do not depend on the proper-class case of [cited theorem 1.1](#) [09]. Source 04 proves the correspondence itself on a class of this kind.

Definition 4.5 (Common positive shifts [04]). A strong k -linear operator L has *common positive shifts* if there is a well-ordered set $S \subseteq \Gamma^{>0}$ with $\text{supp } L(t^g) \subseteq g + S$ for every g . Write S^* for the additive monoid generated by S .

Lemma 4.6 (Iterated operator summability [04]). *If L has common positive shifts, then $(L^n f)_{n \geq 0}$ is Hahn summable for every f , formal power series in L define strong operators, and coefficientwise formal identities between such series hold under composition.*

Proof. $\text{supp } L^n f \subseteq \text{supp } f + nS$. Translate $\text{supp } f$ by its minimum and apply the support principle to the positive well-ordered set $S \cup ((\text{supp } f - \min \text{supp } f) \setminus \{0\})$; a base exponent is used once and later shifts come from S , so every exponent receives finitely many contributions across all paths and all n . For a summable input family use the union of supports; regrouping is justified because every coefficient has finitely many contributions. $\square$

Theorem 4.7 (Derivations with common shifts [04]). *Let $S \subseteq \Gamma^{>0}$ be well ordered and $a_{\delta} : \Gamma \rightarrow k$ additive for $\delta \in S$. Then $D = \sum_{\delta \in S} D_{\delta, a_{\delta}}$ is a strong k -linear derivation with common positive shifts, and every such derivation has a unique expression of this form. When $\bigcap_n n\mathbf{no} = 0$, D maps $\mathcal{R}_o$ into itself if and only if $a_{\delta}(C_{\delta}) = 0$ for all $\delta \in S$.*

Proof. Summability and the Leibniz rule are termwise. Conversely put $a_\delta(g) = [t^{g+\delta}]D(t^g)$; the Leibniz rule for t^{g+h} gives additivity, and strong linearity the expression. If $D(\mathcal{R}_\mathfrak{o}) \subseteq \mathcal{R}_\mathfrak{o}$ and $x \in \Pi$, then $D(x) = nD(x/n) \in n\mathcal{R}_\mathfrak{o}$ for all n , so $D(x) \in \Pi$ by [lemma 2.3](#). For $0 < u \leq \delta$ the coefficient of $D(t^{-u})$ at the nonnegative exponent $\delta - u$ is $a_\delta(-u)$, with no cancellation between distinct shifts, so it vanishes; additivity and divisibility extend this to C_δ . The converse is the no-crossing argument. $\square$

In this theorem D is a derivation; the sources' coefficient ring D is $\mathfrak{o}$ here, which removes the collision 04 flagged.

Theorem 4.8 (Exponential and logarithm on common shifts [04]). *On operators with common positive shifts, $D \mapsto \text{Exp}(D)$ and $\sigma \mapsto \text{Log}(\sigma) = \sum_{n \geq 1} (-1)^{n+1} (\sigma - \text{id})^n / n$ are inverse bijections between strong k -linear derivations and strong k -linear field automorphisms whose difference from the identity has common positive shifts; the shift set may enlarge to $S^* \setminus \{0\}$. They restrict to a bijection between derivations preserving $\mathcal{R}_\mathfrak{o}$ and automorphisms preserving $\mathcal{R}_\mathfrak{o}$ (for $\bigcap_n n\mathfrak{o} = 0$).*

Proof. All series are justified by [lemma 4.6](#). For a derivation D , the iterated Leibniz rule gives multiplicativity of $\text{Exp } D$ after regrouping, and $\text{Exp } D \text{Exp } (-D) = \text{id}$ gives bijectivity; leading terms are fixed since all other shifts are positive. Conversely let $L = \sigma - \text{id}$ and $\sigma^{[X]} = \sum_{n \geq 0} \binom{X}{n} L^n$. For each ordinary $m \geq 0$ this is the binomial expansion of σ^m , so for every exponent γ the t^γ coefficient of $\sigma^{[X]}(fg) - \sigma^{[X]}(f)\sigma^{[X]}(g)$, a polynomial in X by [lemma 4.6](#), vanishes at all $m \geq 0$ and is zero. Differentiating at $X = 0$, with $\frac{d}{dX} \binom{X}{n} \big|_{X=0} = (-1)^{n-1}/n$, shows that $\text{Log } \sigma$ is a derivation; one-variable formal identities give the inverse correspondence. If D preserves $\mathcal{R}_\mathfrak{o}$ it maps Π into Π , so $\text{Exp}(\pm D)$ do; if $\sigma(\Pi) = \Pi$, then $(\sigma - \text{id})(\Pi) \subseteq \Pi$ and the strong logarithmic sum has negative support. $\square$

4.3 Single-shift flows without divisibility

Source 08 treats one flow at a time but allows any ordered abelian group and does not require $\theta(\delta) = 0$. Its statements are translated from the ω -convention ([section 1.5](#)). For an ordered abelian group G , $(k, \mathfrak{o}) = (\mathbb{R}, \mathbb{Z})$ or $(\mathbb{C}, \mathbb{Z}[i])$, let Π_G be the series supported in $G^{<0}$ and $\mathcal{R}_G = \mathfrak{o} \oplus \Pi_G$.

Proposition 4.9 (Falling-factorial iterates [08]). *For any additive $\theta : G \rightarrow k$ and $\delta > 0$, $D_{\delta, \theta}^n(t^\gamma) = \prod_{j=0}^{n-1} (\theta(\gamma) + j\theta(\delta)) t^{\gamma+n\delta}$, and $T^s = \sum_{n \geq 0} s^n D_{\delta, \theta}^n / n!$ ($s \in k$) is a strong k -linear 1-automorphism with $T^s T^u = T^{s+u}$. If $\theta(\delta) = 0$, $T^s(t^\gamma) = t^\gamma \exp(s\theta(\gamma)t^\delta)$; if $c = \theta(\delta) \neq 0$, $T^s(t^\gamma) = t^\gamma (1 + sct^\delta)^{\theta(\gamma)/c}$ as a formal binomial series. In either case $T^s(t^{-\delta}) = t^{-\delta} - s\theta(\delta)$.*

Proof. The iterate formula is induction. Summability, strongness and the group law follow from [lemma 2.1](#) as in [theorem 4.1](#), and the pairs (n, m) with $n + m = r$ are finite in number; multiplicativity follows from the iterated Leibniz rule. The two monomial formulas are the substitutions of the iterate. For $\gamma = -\delta$ every product of length at least two has the factor $\theta(-\delta) + \theta(\delta) = 0$. $\square$

This is `saut:thm:shiftflow` once more, over an arbitrary ordered abelian G ; 08 reproves the summability for that reason.

Theorem 4.10 (Threshold for single flows [08]). *Let G be any ordered abelian group, $\delta > 0$, $\theta : G \rightarrow k$ additive and $s \in k \setminus \{0\}$. The following are equivalent: (a) $T^s(\mathcal{R}_G) \subseteq \mathcal{R}_G$; (b) $T^s(\mathcal{R}_G) = \mathcal{R}_G$; (c) $\theta(C_\delta) = 0$. When they hold, T^s fixes every series supported in C_δ and preserves the constant coefficient of every series.*

Proof. Assume (a). By [lemma 2.3](#), $\Pi_G = \bigcap_n n\mathcal{R}_G$, so $T^s(\Pi_G) \subseteq \Pi_G$. The constant term of $T^s(t^{-\delta})$ is $-s\theta(\delta)$, so $\theta(\delta) = 0$. If $0 > \gamma \in C_\delta$ and $\theta(\gamma) \neq 0$, choose n with $\gamma + n\delta > 0$; the coefficient of $T^s(t^\gamma)$ there is $(s\theta(\gamma))^n/n! \neq 0$, and the exponents $\gamma + m\delta$ are distinct, contradicting $T^s(t^\gamma) \in \Pi_G$. So θ vanishes on the negative part of C_δ , hence on C_δ . Conversely, under (c), monomials with exponents in C_δ are fixed; if $\gamma < 0$ lies outside C_δ , then $\gamma < -n\delta$ for all n , so all $\gamma + n\delta$ stay negative; positive exponents outside C_δ stay positive. So no term enters or leaves exponent 0, $T^{\pm s}(\mathcal{R}_G) \subseteq \mathcal{R}_G$, and (b) follows. (b) $\Rightarrow$ (a) is trivial. $\square$

Only divisibility of *coefficients* by ordinary integers is used, so G need not be divisible. The theorem covers the elementary family only: even at rank one, exponent rescalings and characters preserve the integer part, and it does not assert that every leading-term-preserving automorphism is the exponential of one fixed-shift derivation [08].

Remark 4.11 (Divisibility and the ring $\mathcal{A}_k$ [09]; corrected). Source 09 remarks that divisibility of Γ is essential, because with $\Gamma = \mathbb{Z}$ the translation below preserves $\mathcal{A}_k(\mathbb{Z}) = k[t^{-1}]$. That is right for the boundary ring $\mathcal{A}_k$ and wrong if read for the integer-part ring: the translation $t^{-1} \mapsto t^{-1} - 1$ sends $\frac{1}{2}t^{-1} \in \mathcal{R}_{\mathbb{Z}}(\mathbb{Z}) = \mathbb{Z} + t^{-1}\mathbb{R}[t^{-1}]$ to $\frac{1}{2}t^{-1} - \frac{1}{2} \notin \mathcal{R}_{\mathbb{Z}}(\mathbb{Z})$, as [theorem 4.10](#) predicts. So for $\Gamma = \mathbb{Z}$ the stabilizers of $\mathcal{A}_k$ and of $\mathcal{R}_{\mathbb{Z}}$ differ, and the independence of $\mathfrak{o}$ in [theorem 3.3](#) genuinely needs divisibility, whereas the $\mathcal{R}_{\mathbb{Z}}$ -form of the criterion holds without divisibility for single flows. Whether the general criterion for $\mathcal{R}_{\mathbb{Z}}$ holds for every non-divisible Γ is left open ([question 11.6](#)).

4.4 Why translations fail

Example 4.12 (Translations [04, 08, 09]; the saut example). Let $\Gamma = \mathbb{Q}$, $\delta = 1$, $\theta(\gamma) = \gamma$. Then $D^n(t^\gamma) = \gamma(\gamma+1)\cdots(\gamma+n-1)t^{\gamma+n}$ and $\text{Exp}(D)(t^\gamma) = t^\gamma(1-t)^{-\gamma}$; on t this is $t \mapsto t/(1-t)$, and on t^{-1} it is $t^{-1} \mapsto t^{-1} - 1$, which preserves $k[t^{-1}]$. But

$$\text{Exp}(D)(t^{-1/2}) = t^{-1/2}(1-t)^{1/2} = t^{-1/2} - \frac{1}{2}t^{1/2} - \frac{1}{8}t^{3/2} - \dots$$

has positive support, so neither $\mathcal{A}_k$ nor $\mathcal{R}_{\mathfrak{o}}$ is preserved: $\theta(C_1) \neq 0$. The inverse $t \mapsto t/(1+t)$ is 04's example and $X \mapsto X + s$, $X = t^{-1}$, is 08's. This is exactly the transparent example $T_1(t) = t/(1-t)$ of the surcomplex automorphism report (its $\lambda = \ell$, the constant-coefficient functional, with $\delta = 1$, after `saut:thm:shiftflow`); on the surreals $\ell(1) = 1 \neq 0$, so T_1 does *not* preserve **Oz**: it sends $\omega^{1/2}$ to $\omega^{1/2} - \frac{1}{2}\omega^{-1/2} - \dots \notin \mathbf{Oz}$. Preservation of a few familiar infinite integers is not enough; all rational monomial powers must be tested.

The Taylor coefficient flows of that report (`saut:thm:coeffflow`) move real constants and never preserve **Oz**: this follows from [corollary 5.2](#), and directly, for $d(b) = 1$ they send $bt^{-\delta/2}$ to $bt^{-\delta/2} + t^{\delta/2}$.

4.5 The rank dichotomy and the bottom subgroup

Theorem 4.13 (Rank dichotomy [04, 09]; [08] for single flows). *For a nonzero divisible ordered abelian group Γ , any characteristic-zero k and unital $\mathfrak{o} \subseteq k$: $\mathcal{U}_{\mathcal{R}_{\mathfrak{o}}}(\Gamma) = \{\text{id}\}$ if and only if Γ is Archimedean. For every ordered abelian G , the single flows of [theorem 4.10](#) preserving $\mathcal{R}_G$ are trivial if G is Archimedean, and some are nontrivial if G is a set with a proper C_δ .*

Proof. If Γ is Archimedean, $C_\delta = \Gamma$ for all δ , so by [theorem 3.3](#) every $h_\sigma(g)$ vanishes; σ fixes all monomials, hence everything. If some C_δ is proper, choose (by a basis of the set-sized $\mathbb{Q}$ -space Γ/C_δ , or of the rationalization of the torsion-free G/C_δ for 08's case) a nonzero $\mathbb{Q}$ -linear θ into $\mathbb{Q} \subseteq k$ vanishing on C_δ ; [theorem 4.1](#) or [theorem 4.10](#) gives a nonidentity element. For $\Gamma = \mathbf{No}$ explicit functionals q_β replace the choice ([section 7](#)). $\square$

Definition 4.14 (Bottom convex subgroup [04]). $B(\Gamma) = \bigcap_{\delta > 0} C_\delta$, possibly zero.

Theorem 4.15 (Common fixed field at arbitrary set-sized rank [04]). *For a set-sized nonzero divisible Γ , $\text{Fix}(\mathcal{U}_{\mathcal{R}_0}(\Gamma)) = k((t^{B(\Gamma)}))$.*

Proof. If $g \in B(\Gamma)$, every $a_{\sigma, \delta}(g)$ vanishes by [theorem 3.3](#), so t^g is fixed, and strong linearity fixes the Hahn field on $B(\Gamma)$. If $g \in \text{supp } f$ lies outside $B(\Gamma)$, choose δ with $g \notin C_\delta$ and a $\mathbb{Q}$ -linear θ with $\theta(C_\delta) = 0$ and $\theta(g) = 1$; [theorem 4.2](#) shows Φ_1 moves f . $\square$

Example 4.16 ([04, 09]). Let $\Gamma = \mathbb{Q}^r$, ordered so that the *last* nonzero coordinate decides the sign (09's reverse-lexicographic order; 04 orders by the first coordinate, which is the same group with coordinates reversed). Then the coordinate vectors satisfy $0 < e_1 \ll e_2 \ll \dots \ll e_r$, $C_{e_j} = \mathbb{Q}e_1 \oplus \dots \oplus \mathbb{Q}e_j$, and $B(\Gamma) = \mathbb{Q}e_1$. For $r = 1$ the stabilizer is trivial; for $r \geq 2$ it is nontrivial and fixes exactly $k((t^{\mathbb{Q}e_1}))$. The presence of a smallest Archimedean layer, not the number of layers, controls the fixed field. For $\Gamma = \mathbf{No}$, $B(\mathbf{No}) = 0$: for $g \neq 0$ take $\delta = |g|/\omega$.

4.6 Phase shears

Theorem 4.17 (Finite-dimensional phase shears [08]). *Let H be a convex subgroup of an ordered abelian group G , let $\mathfrak{m}_H$ be the series supported in $H^{>0}$, and let L be an additive map on G vanishing on H , with values in $\mathfrak{m}_H$ and image in a finite-dimensional k -subspace. Then $T_L(t^g) = t^g \exp(L(g))$ extends strongly to a k -linear 1-automorphism of the Hahn field that fixes $k((t^H))$ pointwise, preserves $\mathcal{R}_G$ and constant coefficients, and satisfies $T_L T_M = T_{L+M}$, $T_L^{-1} = T_{-L}$, and $T_L = \text{id} \Leftrightarrow L = 0$.*

Proof. Let $u_1, \dots, u_d$ span the image of L and $U = \bigcup_j \text{supp } u_j$, a well-ordered subset of $H^{>0}$. All supports of all powers in all $\exp(L(g))$ lie in the monoid generated by U , which is well ordered with finite representations; with the input support this gives one summable bound, hence the strong extension. Additivity of L gives multiplicativity. Exponents in H have zero phase, so $k((t^H))$ is fixed; in particular each $M(g)$ and its exponential are fixed, which gives $T_L T_M = T_{L+M}$ on monomials and then strongly. Injectivity follows from injectivity of $\exp$ on $\mathfrak{m}$. If $g \notin H$, every exponent of $t^g \exp(L(g))$ lies in $g + H$, which has constant strict sign by convexity; so positive and negative supports are preserved and no term reaches exponent 0. $\square$

The finite-dimensionality is a sufficient uniform support condition and must not be dropped silently: independent phases for infinitely many exponents need not form a summable family [08]. Taking $H = C_\delta$ and $L(g) = s\theta(g)t^\delta$ recovers the flows of [theorem 4.10](#).

5 The strong stabilizer and all pair automorphisms

Call σ *valued* if $v(\sigma f) = \tau(v(f))$ for an order automorphism τ of Γ . For a character $\chi : (\Gamma, +) \rightarrow (k^\times, \cdot)$ put

$$M_{\chi, \tau} \left(\sum a_\gamma t^\gamma \right) = \sum a_\gamma \chi(\gamma) t^{\tau(\gamma)}. \quad (5.1)$$

It is a strong field automorphism preserving $\mathcal{R}_0$, since τ preserves signs of exponents and $\chi(0) = 1$; $M_{\chi, \tau} M_{\chi', \tau'} = M_{\chi' \cdot (\chi \circ \tau'), \tau \tau'}$ [04, 09].

Theorem 5.1 (Strong valued factorization [04, 09]). *Every strong valued k -fixing automorphism σ with $\sigma(\mathcal{R}_0) = \mathcal{R}_0$ factors uniquely as $\sigma = M_{\chi, \tau} \circ u$ with $u \in \mathcal{U}_{\mathcal{R}_0}(\Gamma)$, where τ is the*

induced exponent automorphism and $\chi(\gamma) = \text{lc}(\sigma(t^\gamma))$. The monomial maps form a subgroup normalizing $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$, and

$$M_{\chi,\tau} D_{\delta,\theta} M_{\chi,\tau}^{-1} = D_{\tau(\delta), \chi(\delta) \theta \circ \tau^{-1}}, \quad (5.2)$$

so the convex condition is invariant, since $\tau(C_\delta) = C_{\tau(\delta)}$.

Proof. Multiplicativity of leading coefficients gives that χ is a character. $u = M_{\chi,\tau}^{-1} \sigma$ fixes the leading term of each monomial; since the images of distinct monomial terms have strictly ordered valuations, strongness shows that u fixes the leading term of every series, and u preserves $\mathcal{R}_o$. The data τ, χ are forced by σ , hence so is u . Conjugating a 1-automorphism by a monomial map gives a 1-automorphism preserving $\mathcal{R}_o$. Formula (5.2) is checked on monomials. $\square$

The ambient decomposition is due to Kuhlmann–Serra [8], and its surcomplex form is the four-layer decomposition `saut:thm:decomp` $U \rtimes \text{Hom}(\Gamma, \mathbb{C}^\times) \rtimes (\text{Aut } \mathbb{C} \times \text{Aut } \Gamma)$ of the surcomplex automorphism report; [theorem 5.1](#) is its restriction to the $\mathcal{R}_o$ -stabilizer with the residue layer trivial. Sources 04 and 09 credit [8]; the new information is the description of u by [theorem 3.3](#). For $k = \mathbb{R}$ and divisible Γ every character is positive, $\chi(\gamma) = \chi(\gamma/2)^2$; for $\Gamma = \mathbb{Q}$ the factor u is absent and every such real automorphism is $t^q \mapsto c^q t^{\lambda q}$ with $c > 0$, $\lambda \in \mathbb{Q}_{>0}$, so the higher-rank shears are a new phenomenon, not complicated one-variable substitutions [09]. The monomial maps include the dilations S_a of the Diophantine report (`odg:def:ex:dilation`) and the character twists ([remark 9.5](#)).

Corollary 5.2 (All pair automorphisms [08, 09]). *Every field automorphism of **No** preserving **Oz** setwise fixes $\mathbb{R}$ pointwise, preserves order and the natural valuation ring, and commutes with P_-, P_0, P_+ and with the floor map of **Oz**; this needs no strongness. Consequently, among strong automorphisms, the **Oz**-stabilizer is exactly $\{M_{\chi,\tau}\} \rtimes \mathcal{U}_{\mathbf{Oz}}$, with χ positive.*

Proof. By `odg:def:thm:autreal` of the Diophantine report, every automorphism of **Oz** extends uniquely to **No**, preserves order, $\mathbb{R}$ and the finite ring, and fixes $\mathbb{R}$ pointwise; restriction and extension are inverse correspondences [08, 09]. The sets and graphs of [theorem 8.2](#) are parameter-free definable, so they are preserved. Fixing $\mathbb{R}$ pointwise makes a strong such map k -fixing, and preserving the valuation ring makes it valued, so [theorem 5.1](#) applies. $\square$

6 Convex rank, solvability, and a wreath product

6.1 The bracket law

Lemma 6.1 (Homogeneous brackets [04, 08, 09]). *For additive $\theta, \psi : \Gamma \rightarrow k$ and $\delta, \epsilon > 0$,*

$$[D_{\delta,\theta}, D_{\epsilon,\psi}] = D_{\delta+\epsilon, \theta(\epsilon)\psi - \psi(\delta)\theta}. \quad (6.1)$$

If $\theta(C_\delta) = 0$ and $\psi(C_\epsilon) = 0$, the bracket is zero when $C_\delta = C_\epsilon$; if $C_\delta \subsetneq C_\epsilon$ it is a multiple of $D_{\delta+\epsilon,\psi}$, whose shift lies in the C_ϵ scale and whose functional vanishes on $C_{\delta+\epsilon} = C_\epsilon$.

Proof. On t^γ the two compositions differ at $\gamma + \delta + \epsilon$ by $\psi(\gamma)\theta(\gamma + \epsilon) - \theta(\gamma)\psi(\gamma + \delta) = \theta(\epsilon)\psi(\gamma) - \psi(\delta)\theta(\gamma)$; strongness extends the identity. Equal scales: each functional kills the other's shift. If $C_\delta \subsetneq C_\epsilon$, then $\psi(\delta) = 0$. $\square$

Source 08 records the same identity in its own normalization (its bracket $[D_{\varphi,\delta}, D_{\psi,\epsilon}] = D_{\psi(\delta)\varphi - \varphi(\epsilon)\psi, \delta+\epsilon}$, which is (6.1) after the sign change of [section 1.5](#)). Elementary derivations with shifts in the same Archimedean class commute; noncommutativity is an interaction

between layers [04]. For general strong derivations the computation applies after expansion into shift coefficients: the two compositions are strongly summable and every output coefficient is a finite sum [09].

6.2 A filtration by convex scales

Say that Γ has *finite ordered rank* r if its distinct nonzero principal convex subgroups are exactly $0 < C_1 < \cdots < C_r = \Gamma$. The layer of a shift δ is the j with $C_\delta = C_j$. A shift in layer r contributes nothing to $\mathfrak{L}_A$ (corollary 3.5), since its functional vanishes on Γ . Let $\mathfrak{F}_j \subseteq \mathfrak{L}_A$ consist of the derivations whose nonzero shifts have layers at least j , so $\mathfrak{L}_A = \mathfrak{F}_1 \supseteq \cdots \supseteq \mathfrak{F}_r = 0$. Selecting the terms of prescribed layers from the images of monomials keeps a summable subfamily, and the new coefficient functions stay additive, so the $\mathfrak{F}_j$ are strongly closed Lie ideals [09].

Lemma 6.2 ([09]). $[\mathfrak{F}_j, \mathfrak{F}_j] \subseteq \mathfrak{F}_{j+1}$ and $[\mathfrak{L}_A, \mathfrak{F}_j] \subseteq \mathfrak{F}_j$.

Proof. Apply lemma 6.1 to pairs of shifts: equal layers cancel and unequal layers give a term in the larger layer. $\square$

Theorem 6.3 (Finite-rank bound [09]). *If the set-sized group Γ has finite ordered rank r , then $\mathfrak{L}_A$ and $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$ have derived length at most $r - 1$.*

Proof. Inductively $\mathfrak{L}_A^{(n)} \subseteq \mathfrak{F}_{n+1}$. For the group, put $G_j = \text{Exp}(\mathfrak{F}_j)$; by the Baker–Campbell–Hausdorff part of cited theorem 1.1 each G_j is a subgroup, and the logarithm of a commutator of two elements of G_j is a strongly summable Lie series each of whose terms contains a bracket of two elements of $\mathfrak{F}_j$, hence lies in $\mathfrak{F}_{j+1}$. So $[G_j, G_j] \subseteq G_{j+1}$, and $G_r = \{\text{id}\}$. $\square$

The Lie bound is elementary; the group bound uses the imported correspondence, which [1] proves for set-sized Γ .

6.3 Sharpness, including the abstract group

In $\mathbb{Q}^r$ ordered as in example 4.16, let θ_i extract the i th coordinate, $X_{j,i} = D_{e_j, \theta_i}$ for $j < i$, and

$$a_0 = 0, \quad a_k = \sum_{h=1}^k 2^{k-h} e_h, \quad Y_{k;j,i} = D_{a_k + e_j, \theta_i} \quad (k < j < i \leq r).$$

These satisfy the criterion, since $C_{e_j} = \mathbb{Q}e_1 \oplus \cdots \oplus \mathbb{Q}e_j$.

Lemma 6.4 (Balanced brackets [09]). $Y_{k;j,i} \in \mathfrak{L}_A^{(k)}$, and $[Y_{k-1;k,j}, Y_{k-1;j,i}] = Y_{k;j,i}$ for $k \geq 1$.

Proof. At $k = 0$ this is the definition. In the bracket, $\theta_j(a_{k-1} + e_j) = 1$ and $\theta_i(a_{k-1} + e_k) = 0$, and the shift is $2a_{k-1} + e_k + e_j = a_k + e_j$; apply (6.1) and induct. $\square$

Lemma 6.5 (Mixed-parameter detection [09]). *Take finitely many homogeneous positive-shift derivations and an iterated Lie bracket W of them. Replace each leaf occurrence D_ℓ by $\text{Exp}(s_\ell D_\ell)$ with independent scalar parameters and each bracket by the group word $ghg^{-1}h^{-1}$, giving $\mathcal{W}(\mathbf{s})$. In its action on a monomial, the coefficient of the product of all leaf parameters is the action of W . If $W \neq 0$, some specialization $\mathbf{s} \in k^m$ gives a nonidentity group word.*

Proof. Formally, $(1+sD+\cdots)(1+uE+\cdots)(1-sD+\cdots)(1-uE+\cdots) = 1+su[D, E] + \text{terms of other multidegrees}$; induct on the bracket tree, using that the subtrees have disjoint parameter sets and that a group word is the identity when a required leaf parameter is zero. Choose a monomial and an output exponent on which W acts nontrivially. Only a finite set of shifts occurs, so by the support principle the coefficient at that output is a polynomial in the parameters, nonzero by the first part. An infinite field has a point where it does not vanish. Repeated leaves are handled by their separate parameters. $\square$

Theorem 6.6 (Sharp derived length [09]). *For $\Gamma = \mathbb{Q}^r$ as above, every characteristic-zero k and every unital $\mathfrak{o} \subseteq k$, the group $\mathcal{U}_{\mathcal{R}_\mathfrak{o}}(\Gamma)$ and its Lie algebra $\mathfrak{L}_A$ have derived length exactly $r - 1$.*

Proof. For $r = 1$ use [theorem 4.13](#). For $r \geq 2$, $0 \neq Y_{r-2;r-1,r} \in \mathfrak{L}_A^{(r-2)}$ by [lemma 6.4](#), and [lemma 6.5](#) turns its balanced bracket into a nonidentity element of the $(r - 2)$ nd derived subgroup. The upper bound is [theorem 6.3](#). $\square$

The lower bound does not identify the abstract derived subgroup with the exponential of the derived Lie algebra; it uses independent parameters and a nonzero mixed coefficient. The upper bound uses BCH and strongly closed ideals [09].

6.4 Metabelian and not nilpotent: one group, printed once

On $\Gamma = \mathbf{No}$ put $D = D_{1,q_1}$, $E = D_{\omega,q_2}$ and $E_n = D_{\omega+n,q_2}$. Both D and E satisfy the criterion: q_1 kills the finite exponents and q_2 kills every exponent bounded by an ordinary multiple of ω .

Proposition 6.7 ([04, 09]). *$[D, E_n] = E_{n+1}$ and $[E_m, E_n] = 0$; the E_n are nonzero and linearly independent over $\mathbb{R}$. So the Lie algebra spanned by $D, E_0, E_1, \dots$ is metabelian and not nilpotent. The same holds in rank three: in $\mathbb{Q}^3$ with $X = D_{e_1,\theta_2}$, $Y = D_{e_2,\theta_3}$ one has $(\text{ad } X)^n Y = D_{e_2+n e_1,\theta_3} \neq 0$.*

Proof. $q_1(\omega+n) = 1$, $q_2(1) = q_2(\omega+n) = 0$; apply (6.1). $E_n(t^{\omega^2}) = t^{\omega^2+\omega+n}$ gives independence. The rank-three case is the same computation [09]. $\square$

Put $A_s = \text{Exp}(sD)$ and $B_u = \text{Exp}(uE)$ for $s, u \in \mathbb{R}$, and $[F, G] = FGF^{-1}G^{-1}$. Here e^{st} denotes the Hahn series $\exp(st)$ in the monomial $t = \omega^{-1}$.

Theorem 6.8 (Exact commutators [04, 08]). *$A_s B_u A_s^{-1} = \text{Exp}(u e^{st} E)$ and $[A_s, B_u] = \text{Exp}(u(e^{st} - 1)E)$. With $C_0 = B_u$ and $C_{n+1} = [A_s, C_n]$,*

$$C_n = \text{Exp}(u(e^{st} - 1)^n E), \quad C_n(t^{-\omega^2}) = t^{-\omega^2} \exp(-u(e^{st} - 1)^n t^\omega),$$

so for $s, u \neq 0$, $C_n(t^{-\omega^2}) - t^{-\omega^2}$ has leading term $-us^n t^{-\omega^2+\omega+n}$. Hence $\langle A_s, B_u \rangle$ is metabelian and not nilpotent. At $s = u = -1$, the leading term is $(-1)^n \omega^{\omega^2-\omega-n}$.

Proof. D fixes t (as $q_1(1) = 0$) and $A_s(t^\omega) = t^\omega e^{st}$. On monomials, $A_s E A_s^{-1}(t^\gamma) = q_2(\gamma) t^{\gamma+\omega} e^{st}$, because E kills every series in t and $q_1(\gamma + \omega) - q_1(\gamma) = 1$; so $A_s E A_s^{-1} = e^{st} E$, and exponentiating gives the first formula. E kills $\mathbb{R}[[t]]$, so the derivations $h(t)E$, $h \in \mathbb{R}[[t]]$, commute and their exponentials add coefficients, which gives the second formula and, by induction, C_n . All shifts lie in the monoid generated by the well-ordered set $\{1\} \cup (\omega + \mathbb{N})$, so [theorem 4.8](#) justifies all sums. The maps $\text{Exp}(h(t)E)$ form an abelian group normalized by A_s , which acts by $h \mapsto e^{st}h$; so the generated group is abelian-by-abelian. Since $q_2(-\omega^2) = -1$, $C_n(t^{-\omega^2})$ is as displayed, and $(e^{st} - 1)^n$ has leading term $s^n t^n$; the leading term of the difference is therefore nonzero at every depth, so the lower central series never terminates. $\square$

Source 08 works with $A_1^{(08)} = A_{-1}$ and $B_1^{(08)} = B_{-1}$ and proves faithfulness, which identifies the group. Write $v = t = \omega^{-1}$, $u_0 = t^\omega = \omega^{-\omega}$, $P = t^{-\omega} = \omega^\omega$, $M = t^{-\omega^2} = \omega^{\omega^2}$, and for h supported in $C_\omega^{>0}$ let $N_h(t^\gamma) = t^\gamma \exp(-q_2(\gamma)h)$, a phase shear ([theorem 4.17](#)) with $N_h N_{h'} = N_{h+h'}$, fixing v, u_0, P and with $N_h(M) = M \exp(h)$. Then $B_{-1} = N_{u_0}$, $A_{-1}(u_0) = u_0 e^{-v}$, $A_{-1}(P) = P e^v$, and $A_{-1} N_h A_{-1}^{-1} = N_{A_{-1}(h)}$.

Lemma 6.9 (Finite exponential independence [08]). *For a Laurent polynomial $p \in \mathbb{Z}[X, X^{-1}]$ put $h_p = u_0 p(e^{-v})$ and $\mathcal{N}_p = N_{h_p}$. Then $p \mapsto p(e^{-v})$ is injective, $p \mapsto \mathcal{N}_p$ is an injective homomorphism, and $A_{-1} \mathcal{N}_p A_{-1}^{-1} = \mathcal{N}_{Xp}$.*

Proof. If $\sum_{n \in F} k_n e^{-nv} = 0$ with $|F| = r$ distinct integers, comparing the coefficients of v^j , $0 \leq j < r$, gives a Vandermonde system with distinct nodes $-n$, so all $k_n = 0$. Injectivity of $\mathcal{N}$ follows from $\mathcal{N}_p(M) = M \exp(h_p)$, and $A_{-1}(h_p) = u_0 e^{-v} p(e^{-v}) = h_{Xp}$. $\square$

Theorem 6.10 (A faithful $\mathbb{Z} \wr \mathbb{Z}$ [08]). *The map $\mathbb{Z}[X, X^{-1}] \rtimes \mathbb{Z} \rightarrow \mathcal{U}_{\mathbf{Oz}}$, $(p, m) \mapsto \mathcal{N}_p A_{-1}^m$, with $(p, m)(q, n) = (p + X^m q, m + n)$, is an injective homomorphism onto $\langle A_{-1}, B_{-1} \rangle$. So this group is the restricted wreath product $\mathbb{Z} \wr \mathbb{Z}$, of derived length exactly two. It extends $\mathbb{C}$ -linearly to $\mathbf{No}(i)$ preserving $\mathbf{Oz}[i]$, and it already lives on the set-sized exponent group $\mathbb{Q}\omega^2 + \mathbb{Q}\omega + \mathbb{Q}$.*

Proof. [Lemma 6.9](#) gives the multiplication law, and $\mathcal{N}_{X^n} = A_{-1}^n B_{-1} A_{-1}^{-n}$ shows that the image is generated by A_{-1}, B_{-1} . If $\mathcal{N}_p A_{-1}^m = \text{id}$, applying it to P gives $P e^{mv} = P$, so $m = 0$, and then to M gives $h_p = 0$, so $p = 0$. Finite Laurent polynomials under multiplication by X are $\bigoplus_{\mathbb{Z}} \mathbb{Z}$ with the shift. The group is nonabelian and its normal subgroup $\mathcal{N}$ is abelian with cyclic quotient. $\square$

Remark 6.11 (All nonzero times [merge]). The same proof gives $\langle A_s, B_u \rangle \cong \mathbb{Z} \wr \mathbb{Z}$ for all real $s, u \neq 0$: here $B_u = N_{-ut^\omega}$, $A_s N_h A_s^{-1} = N_{A_s(h)}$ and $A_s(t^\omega) = t^\omega e^{st}$, so with $h_p = -ut^\omega p(e^{st})$ the map $(p, m) \mapsto N_{h_p} A_s^m$ obeys the same law; the Vandermonde nodes ns are distinct, and $A_s^m(P) = P e^{-mst}$ forces $m = 0$.

Corollary 6.12 (Every set stabilizer is not nilpotent [04, 08]). *For every set $P \subseteq \mathbf{No}$, the pointwise stabilizer of P in $\mathcal{U}_{\mathbf{Oz}}$ contains a subgroup isomorphic to $\mathbb{Z} \wr \mathbb{Z}$, metabelian and not nilpotent.*

Proof. Choose ordinals $0 < \beta < \beta'$ above every inner exponent of every outer exponent of every member of P ([lemma 7.1](#)), and replace ω, ω^2 by $\epsilon = \omega^\beta, \omega^{\beta'}$, using the functionals $q_\beta, q_{\beta'}$ and the shifts $1, \epsilon$. Both functionals vanish on the exponents of P and satisfy the criterion, and $q_\beta(\epsilon) = 1, q_\beta(1) = q_{\beta'}(\epsilon) = q_{\beta'}(1) = 0$; the proofs are unchanged. $\square$

These examples live in rank three, consistent with [theorem 6.3](#) (derived length at most two there) [08, 09].

7 Fixed fields and arbitrarily large stabilizers

7.1 Exponent coefficient functionals

For $\beta \in \mathbf{No}$ let $q_\beta(\gamma) = [\omega^\beta]\gamma$ be the coefficient of ω^β in the Conway normal form of the exponent γ (the *inner* normal form; the outer support of $f = \sum c_\gamma t^\gamma$ consists of the exponents γ). Normal-form addition makes $q_\beta : \mathbf{No} \rightarrow \mathbb{R}$ additive; it need not commute with infinite sums inside the exponent group, and only finite additivity is used [04]. If $0 < \delta \ll \omega^\beta$ then $q_\beta(C_\delta) = 0$: a nonzero γ with $|\gamma| \leq n\delta$ has leading Conway exponent at most that of δ , which is below β [04, 08, 09].

Lemma 7.1 (Parameter avoidance [04, 08, 09]). *Let $P \subseteq \mathbf{No}(i)$ be a set, E the union of the outer supports of the real and imaginary parts of its members, and T the union of the inner supports of the members of E . Then E and T are sets, and for every ordinal $\beta > 0$ above all members of T , q_β vanishes on E .*

Proof. Replacement and union; every set of surreals has a strict upper bound, for example an ordinal above all their birthdays. This bounds a set of supports; it does not choose a bound for all exponents at once. $\square$

Theorem 7.2 (The common fixed field [04, 08, 09]). *$\text{Fix}(\mathcal{U}_{\mathbf{Oz}}) = \mathbb{R}$, and already the elementary shears σ_{δ, q_β} have common fixed field $\mathbb{R}$. The same holds for all field automorphisms of $\mathbf{No}$ preserving $\mathbf{Oz}$. The common fixed elements in $\mathbf{Oz}$ are $\mathbb{Z}$, and the common fixed subring of all ring automorphisms of $\mathbf{Oz}$ is $\mathbb{Z}$.*

Proof. All these maps fix $\mathbb{R}$ (corollary 5.2). If $f \notin \mathbb{R}$, some nonzero $\gamma \in \text{supp } f$ has $q_\beta(\gamma) \neq 0$ for some β , for instance its leading inner exponent [08]. Put $\theta = q_\beta$ and $\delta = \omega^{\beta-1}$; then $\theta(C_\delta) = 0$ and $\theta(\delta) = 0$, and theorem 4.2 shows that $\sigma_{\delta, \theta}$ moves f : its first displacement is $a_\lambda \theta(\lambda) t^{\lambda+\delta}$, λ the least exponent of f with $\theta(\lambda) \neq 0$. Ring automorphisms of $\mathbf{Oz}$ are restrictions of field automorphisms preserving $\mathbf{Oz}$ (odg: def: thm: autreal). $\square$

Source 04 reaches the same conclusion through lemma 7.3 with $V = 0$; the explicit choice above is 08's and 09's.

7.2 Relative fixed fields and the Hahn hull

Lemma 7.3 (Set-sized coefficient separation [04]). *Let $V \subseteq \mathbf{No}$ be a set-sized $\mathbb{Q}$ -subspace and $g \in \mathbf{No} \setminus V$. There are $\delta > 0$ and an additive class map $a : \mathbf{No} \rightarrow \mathbb{R}$ with $a(V) = 0$, $a(C_\delta) = 0$ and $a(g) = 1$.*

Proof. Let $T = \bigcup_{x \in V \cup \{g\}} \text{supp}_\omega(x)$, a nonempty set of inner exponents, and $p_T(x) = ([\omega^\beta]x)_{\beta \in T} \in \mathbb{R}^T$. It is injective on the set-sized space $W = V + \mathbb{Q}g$, so $\ell_0(p_T(v + qg)) = q$ is a well-defined $\mathbb{Q}$ -linear functional on $p_T(W)$; extend it through a basis of the set $\mathbb{R}^T$ to ℓ and put $a = \ell \circ p_T$. Choose β_0 below every member of T and $\delta = \omega^{\beta_0}$; then $p_T(C_\delta) = 0$. $\square$

For a set $A \subseteq k((t^{\mathbf{No}}))$ ($k = \mathbb{R}$ or $\mathbb{C}$) let S_A be the union of the outer supports, $V_A = \text{span}_{\mathbb{Q}}(S_A)$ and $\mathcal{H}(A) = k((t^{V_A}))$; let $\mathcal{E}_A$ be the group generated by the elementary shears Φ_s (with real s) that fix A pointwise, in the sense of section 1.4.

Theorem 7.4 (Exact elementary hull [04]). *For every set $A \subseteq k((t^{\mathbf{No}}))$, $\text{Fix}(\mathcal{E}_A) = \mathcal{H}(A) = k((t^{V_A}))$.*

Proof. By theorem 4.2 a nonidentity shear with functional a fixes all of A iff a vanishes on S_A , iff $a(V_A) = 0$; such shears fix every series supported in V_A . If $f \notin \mathcal{H}(A)$, take $g \in \text{supp } f \setminus V_A$ and apply lemma 7.3. $\square$

Corollary 7.5 (Monomial parameters [04]). *For a set $S \subseteq \mathbf{No}$ with $V = \text{span}_{\mathbb{Q}}(S)$, the pointwise stabilizer in $\mathcal{U}_{\mathbf{Oz}}$ of $\{t^g : g \in S\}$ has common fixed field exactly $\mathbb{R}((t^V))$.*

Proof. The elementary subgroup already has this fixed field. Conversely an ordered field automorphism fixing t^g fixes its positive rational powers, hence all t^v , $v \in V$, and strong linearity fixes every series supported on V . $\square$

Naming $\omega = t^{-1}$ gives the fixed field $\mathbb{R}((t^{\mathbb{Q}}))$, much larger than $\mathbb{R}(\omega)$; naming all monomials with real exponents gives $\mathbb{R}((t^{\mathbb{R}}))$ [04]. On sets, $A \mapsto \mathcal{H}(A)$ is extensive, monotone and idempotent, and $\mathcal{H}(A)$ is the smallest Hahn subfield with divisible exponent group containing every monomial of the normal forms of A [04]. For nonmonomial parameters [theorem 7.4](#) computes the fixed field of $\mathcal{E}_A$ only: a general strong automorphism fixing a series may do so by cancellation, and $\mathcal{H}(A)$ is not asserted to be the model-theoretic definable closure [04] ([question 11.3](#)).

7.3 Product actions and nonsolvable stabilizers

Theorem 7.6 (Faithful $\mathbb{R}^\kappa$ actions [09]). *For every set $P \subseteq \mathbf{No}$ and every set cardinal κ there is a faithful action of the full product $\mathbb{R}^\kappa$ by strong 1-automorphisms preserving $\mathbf{Oz}$ and fixing P pointwise.*

Proof. Let E, T be as in [lemma 7.1](#), and choose an ordinal β above κ and all of T . For $j < \kappa$ put $\delta_j = \omega^j$, $\eta_j = \omega^{\beta+j}$ (ordinal addition in the index) and $\theta_j = q_{\beta+j}$. Then $\theta_j|_E = 0$, $\theta_j(C_{\delta_j}) = 0$, $\theta_j(\delta_\ell) = 0$ and $\theta_j(\eta_\ell)$ is the Kronecker delta. For $\mathbf{a} \in \mathbb{R}^\kappa$ let $D_{\mathbf{a}}(t^\gamma) = t^\gamma \sum_{j < \kappa} a_j \theta_j(\gamma) t^{\delta_j}$; the shifts form one well-ordered set, so [proposition 4.4](#) applies with no finiteness condition on $\mathbf{a}$. $D_{\mathbf{a}}$ kills the exponents of P . The $D_{\mathbf{a}}$ commute by (6.1) and add, so $\mathbf{a} \mapsto \text{Exp}(D_{\mathbf{a}})$ is a homomorphism, and $\text{Exp}(D_{\mathbf{a}})(t^{\eta_j}) = t^{\eta_j} \exp(a_j t^{\delta_j})$ detects every a_j . The result is one class relation encoding a set-indexed action. $\square$

Theorem 7.7 (Nonsolvable stabilizers [09]). *For every set $P \subseteq \mathbf{No}$ the pointwise stabilizer of P in $\mathcal{U}_{\mathbf{Oz}}$ is nonsolvable; for every $r \geq 2$ it contains a subgroup whose derived length is at least $r - 1$, and exactly $r - 1$ granted [cited theorem 1.1](#) for $\Gamma = \mathbf{No}$.*

Proof. Choose ordinals $\beta \leq \eta_1 < \dots < \eta_r$ above T , and put $e_j = \omega^{\eta_j}$, $\theta_i = q_{\eta_i}$. Then $e_1 \ll \dots \ll e_r$, $\theta_i(e_j)$ is the Kronecker delta, and each D_{e_j, θ_i} ($j < i$) satisfies the criterion and kills P . The balanced brackets of [lemma 6.4](#) are unchanged, so [lemma 6.5](#) shows that the group generated by all $\text{Exp}(s D_{e_j, \theta_i})$, $s \in \mathbb{R}$, has derived length at least $r - 1$; this already gives nonsolvability. The upper bound applies the filtration by the $r - 1$ scales $C_{e_1}, \dots, C_{e_{r-1}}$ as in [theorem 6.3](#); its group step uses BCH. $\square$

8 What the pair (No, Oz) remembers

Sources 04, 08 and 09 each reconstruct the coefficient field from $\mathbf{Oz}$; 08 and 09 add first-order definitions of the ordinary integers and of Π . All of this is now in the omnific Diophantine report, placed after the three pins, and is cited rather than reprinted:

- Π is defined in $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ by $\exists y (x^2 = 2y^2)$ (`odg:def:thm:ideal`); the first-order definitions of Π and of the graph of ct through ordinary integers are `odg:cor:definablect`, and a positive-existential definition of $\mathbb{Z}$ is `odg:thm:guard`; Pell rigidity is `odg:cor:pell`;
- the multiplier identity $(\Pi : \Pi) = \mathcal{A}_k$, $k = \{0\} \cup \mathcal{A}_k^\times$ and its internal first-order form (`odg:def:thm:multiplier`, `odg:def:cor:internal`); $\text{Frac } \mathbf{Oz} = \mathbf{No}$ (`odg:thm:fractions`);
- the pure ring $\mathbf{Oz}$ interprets $\mathbf{No}$ with $\mathbb{R}$, the order, the finite ring, the infinitesimals and the standard part (`odg:def:thm:realrecovery`); the value group (`odg:def:cor:valuegroup`); every automorphism of $\mathbf{Oz}$ extends uniquely to $\mathbf{No}$ and fixes $\mathbb{R}$ pointwise (`odg:def:thm:autreal`);

- for $\mathbf{Oz}[i]$: $\mathbb{C}$ is definable (`odg:def:cor:internal`), every field automorphism of $\mathbb{C}$ lifts coefficientwise and restriction to $\mathbb{C}$ splits (`odg:def:prop:coeffaut`); a phase twist shows that neither $\mathbf{No}$ nor conjugation is definable from $\mathbf{Oz}[i]$ (`odg:def:lem:twist`, `odg:def:thm:norealaxis`).

In the sources: 04's intrinsic coefficient invariance, 08's characteristic reconstruction, its restriction correspondence and its split restriction to $\text{Aut } \mathbb{C}$, and 09's corollary on all pair automorphisms re-derive `odg:def:thm:multiplier`, `odg:def:thm:autreal` and `odg:def:prop:coeffaut`. 08's formula $J(x) :\Leftrightarrow \forall n (\text{Int}(n) \wedge n \neq 0 \Rightarrow \exists y x = ny)$ is the definition `odg:cor:definablect` in structure (present at 08's pin); 09's order-free formula $\Phi_J(a) :\Leftrightarrow a \in \mathbf{Oz} \wedge \forall u, v \in \mathbf{Oz} [u^2 - 2v^2 = 1 \Rightarrow \exists b \in \mathbf{Oz} a = ub]$ is a variant using all Pell first coordinates; both are dominated by the one-witness existential formula `odg:def:thm:ideal`, and 08's existential definition of $\mathbb{Z}$ (six witnesses, two inequations) by `odg:thm:guard`. 09 observes that the order symbol can be omitted, positivity being definable by nonzero squares. The following extras are not in the Diophantine report.

Lemma 8.1 (Order on $\mathbf{Oz}$ through fractions [08]). *For $z \in \mathbf{Oz}$, $z \geq 0$ if and only if $\exists a, b \in \mathbf{Oz} (b \neq 0 \wedge a^2 = zb^2)$.*

Proof. If $a^2 = zb^2$ with $b \neq 0$ then $z = (a/b)^2 \geq 0$. Conversely a square root of $z \geq 0$ exists in the real closed field $\mathbf{No}$ and is a quotient a/b of omnific integers (`odg:thm:fractions`). $\square$

The formula is existential with one inequation, consistent with the Diophantine report's result that the nonnegative cone is not positive-existentially definable (`odg:prop:notdiophantine`).

Theorem 8.2 (Zero-cut projections and floor [08, 09]). *In $(\mathbf{No}, +, \cdot, <, \mathbf{Oz})$, equivalently in the fraction field interpreted in the pure ring $\mathbf{Oz}$, the graphs of the projections P_-, P_0, P_+ on all of $\mathbf{No}$ (so $P_0 = \text{ct}$ extended to $\mathbf{No}$) and of the floor map $x \mapsto \lfloor x \rfloor_{\mathbf{Oz}}$ are parameter-free definable. For every monomial $m = t^\alpha$, truncation below α , $T_{<\alpha}(f) = m P_-(f/m)$, is definable with the single parameter m . The same definitions work in $(\mathbb{R}((t^\Gamma)), \mathcal{R}_{\mathbb{Z}})$ for every nonzero divisible ordered abelian Γ .*

Proof. $\Pi, \mathbb{R}, \mathcal{O}$ and $\mathfrak{m}$ are definable by the results cited above (in the t -convention $\mathcal{O}$ is the finite ring and $\mathfrak{m}$ the infinitesimals). For $x \in \mathbf{No}$, $P_-(x)$ is the unique $j \in \Pi$ with $x - j \in \mathcal{O}$: existence by splitting the support at 0, uniqueness from $\Pi \cap \mathcal{O} = 0$. Then $P_0(x)$ is the unique $c \in \mathbb{R}$ with $x - j - c \in \mathfrak{m}$, since $\mathbb{R} \cap \mathfrak{m} = 0$, and $P_+ = \text{id} - P_- - P_0$ [09]. The floor graph is $y \in \mathbf{Oz} \wedge y \leq x < y + 1$, unique because distinct omnific integers differ by at least one [08]; its value is `odg:thm:floor`. Dividing by m translates exponents by $-\alpha$, P_- keeps those below 0, and multiplying by m restores them [09]. For the set-sized field the proof uses only real coefficients, the Hahn order (the field is real closed since Γ is divisible), and the definitions of Π and $\mathbb{R}$: the formula of `odg:def:thm:ideal` works for every Γ , and the multiplier identity `odg:def:thm:multiplier` holds inside the whole Hahn field by the same one-monomial proof [09]. $\square$

Every pair automorphism therefore preserves $\Pi \oplus \mathbb{R} \oplus \mathfrak{m}$ summandwise, fixes the real summand, and satisfies $\text{ct}(\sigma x) = \text{ct}(x)$ and $\sigma(\lfloor x \rfloor) = \lfloor \sigma x \rfloor$ ([corollary 5.2](#)).

Remark 8.3 (Zero truncation is not the normal form [08, 09]). P_0 is definable on all of $\mathbf{No}$ but is not multiplicative there. Definability of P_- does not define the set of monomials, and truncation at a named monomial is not a uniform definable choice of such a boundary at every valuation; the distinction matters in comparison with work on truncation [2]. Recovering every coefficient by $[\omega^g]x = \text{ct}(\omega^{-g}x)$ would need the monomial map $g \mapsto \omega^g$, and [section 9](#) shows that this is a genuine obstruction.

Source 08's Question 2, whether the reconstruction of $\mathbb{C}$ from the pure ring $\mathbf{Oz}[i]$ is first-order, is answered positively by `odg:def:cor:internal`, whose formula $\exists y (x^2 = 2y^2)$ defines $\Pi_{\mathbb{C}}$ in $\mathbf{Oz}[i]$ and whose multiplier formula then defines $\mathbb{C}$; 09 gives an independent route, its Pell-divisibility formula with the Gaussian Pell solutions (such as (i, i)) handled, followed by the idealizer. The question is not repeated here.

9 What the pair forgets: monomials and representatives

Theorem 9.1 (Set-parameter invisibility [04, 08, 09]). *For every set $P \subseteq \mathbf{No}$ there is a nonidentity strong real-linear 1-automorphism σ of $\mathbf{No}$ with $\sigma(\mathbf{Oz}) = \mathbf{Oz}$ that fixes $P \cup \mathbb{R} \cup \{\omega\}$ pointwise and sends a positive Conway monomial in $\mathbf{Oz}$ to a nonmonomial in $\mathbf{Oz}$. Explicitly, with β as in lemma 7.1 and $\eta = \omega^\beta$, one may take $\sigma = \sigma_{1,q_\beta}$, and then*

$$\sigma(t^{-\eta}) = t^{-\eta} \exp(-t) = \sum_{n \geq 0} \frac{(-1)^n}{n!} t^{-\eta+n}, \quad \text{that is} \quad \sigma(\omega^{\omega^\beta}) = \sum_{n \geq 0} \frac{(-1)^n}{n!} \omega^{\omega^\beta-n}. \quad (9.1)$$

Consequently, in $\mathfrak{S} = (\mathbf{No}, 0, 1, +, \cdot, <, \mathbf{Oz})$ neither the monomial class $\mathfrak{M} = \{\omega^x : x \in \mathbf{No}\}$ nor the graph of $x \mapsto \omega^x$ is definable with parameters from any set, even after naming $\mathbb{R}$ as a predicate, every ordinary real, ω , and any further set of surreal constants.

Proof. q_β kills C_1 since $\beta > 0$, so $\sigma \in \mathcal{U}_{\mathbf{Oz}}$ by theorem 4.1. It fixes each monomial with exponent in E , hence every member of P by strongness, and it fixes $\mathbb{R}$ and $t^{-1} = \omega$ because $q_\beta(-1) = 0$. Since $q_\beta(-\eta) = -1$, (9.1) follows; all exponents $-\eta + n$ are negative because η is infinite, and the terms at $-\eta$ and $-\eta + 1$ show that the image is not a monomial. A class definable from parameters fixed pointwise by an automorphism of $\mathfrak{S}$ is invariant under it; σ preserves every symbol and moves an element of $\mathfrak{M}$ out of $\mathfrak{M}$. A definable graph of the omega-map would make its range definable by an existential quantifier. $\square$

The three sources give the same construction: 04 with $a = [\omega^\beta]$ and explicit fixing of $\mathbb{R} \cup \{\omega\}$, 08 with the witness $\omega^{\omega^\alpha} \mapsto \sum_n \omega^{\omega^\alpha-n}/n!$ (its sign convention), 09 with q_η . An individual formula uses finitely many parameters; the theorem is the stronger statement that one automorphism fixes an entire prescribed set. Nothing is claimed after naming a proper class of parameters: naming every surreal would remove the symmetry [04, 08]. Expanding the structure by the already definable order, floor or zero-truncation maps cannot remove the obstruction [08].

Corollary 9.2 ([04, 09]). *The monomial class is not determined, as an automorphism-invariant subclass, by $\mathbf{Oz}$, its ordered fraction field and any set of parameters; the standard normal-form section of the natural valuation is not first-order recoverable from these data. The Conway monomials lying in $\mathbf{Oz}$ are not definable in the pure ring $\mathbf{Oz}$ with parameters from any set.*

Proof. The range of the section is $\mathfrak{M}$. For the pure ring, restrict the witness of theorem 9.1 to $\mathbf{Oz}$, with parameters in $\mathbf{Oz}$; automorphisms of $\mathbf{Oz}$ and field automorphisms of $\mathbf{No}$ preserving $\mathbf{Oz}$ correspond (`odg:def:thm:autreal`). $\square$

9.1 No definable representative rule

Write $x \sim_v y$ if $v(x) = v(y)$ on $\mathbf{No}^\times$. A *valuation representative rule* is a function $s : \mathbf{No}^\times \rightarrow \mathbf{No}^\times$ with $v(s(x)) = v(x)$ and $s(x) = s(y)$ whenever $x \sim_v y$. For a set-sized field this is a section of the quotient by $\mathcal{O}^\times$; the formulation also makes sense on the class $\mathbf{No}$ without treating proper-class equivalence classes as elements. No multiplicativity, positivity or monomial-image assumption is imposed [09].

Lemma 9.3 (A whole fiber moves [09]). *For σ and η as in [theorem 9.1](#), every f with $v(f) = -\eta$ satisfies $v(\sigma f - f) = -\eta + 1$ and $\text{lc}(\sigma f - f) = -\text{lc}(f)$.*

Proof. This is [theorem 4.2](#) with $\gamma_0 = -\eta$ and $\theta(-\eta) = -1$: later input terms contribute beyond $-\eta + 1$, higher iterates at $-\eta + n$, $n \geq 2$. $\square$

Theorem 9.4 (No definable section [09]). *For every set $P \subseteq \mathbf{No}$, no P -definable function on $\mathbf{No}^\times$ is a valuation representative rule.*

Proof. Let $x = t^{-\eta}$. Since σ fixes valuations, $\sigma x \sim_v x$ and $s(\sigma x) = s(x)$, while P -definability gives $s(\sigma x) = \sigma(s(x))$. So $s(x)$ would be a fixed point of σ with valuation $-\eta$, contradicting [lemma 9.3](#). $\square$

This is the strongest nondefinability statement of the sources; it covers any representative rule, including one-term ones, and does not follow from character twists. The nondefinability is relative to the field-and-integer-part language; Conway's normal forms remain set-theoretically defined in the ambient foundation [09].

Remark 9.5 (Character twists, one-term series and the RV sort [04, 02]). Bare nondefinability of the coefficient-one monomials already follows from character twists: for a nonzero additive $a : \mathbf{No} \rightarrow \mathbb{R}$ with $a(V_A) = 0$ ([lemma 7.3](#)), $\sum c_g t^g \mapsto \sum c_g e^{a(g)} t^g$ preserves $\mathbf{Oz}$, fixes A and sends some monomial to a non-unit multiple of itself [04]. These twists are the $M_{\chi, \text{id}}$ of [theorem 5.1](#); they are 02's $\tau_{\chi, t}$ ([remark 17.10](#)), the real form of the Diophantine report's phase twist (`odg:def:lem:twist`), and the collection's character twists (`saut:thm:decomp`); they are established and not claimed new. What the shears add is motion inside the leading-term kernel with more than one nonzero term. In particular the class of nonzero one-term series $\{c\omega^g : c \in \mathbb{R}^\times, g \in \mathbf{No}\}$ is not definable from the ordered field, $\mathbf{Oz}$ and set parameters: the shear sends a one-term series to a multiterm one, which rescaling cannot do [04]. Nondefinability also survives adding the sort $\text{RV}(K) = K^\times / (1 + \mathfrak{m})$ with its multiplication, the quotient map and every element of that sort named, since a 1-automorphism acts trivially on it (for the class $\mathbf{No}$ this is a metatheoretic expansion, read formula by formula). This must be distinguished from naming a representative map $\text{RV}(K) \rightarrow K$, which the Conway monomials partly constitute and which would destroy the symmetry by design: $\Phi_s(t^{-\omega})$ has leading term $t^{-\omega}$, yet Φ_s moves $t^{-\omega}$ [04].

Remark 9.6 (Contrasts in the collection). The single dilation $S_2(\sum a_g t^g) = \sum a_g t^{2g}$ defines the canonical monomials, every coefficient and truncation in the language of fields with one automorphism (`dsup:thm:definability` in `docs/surcomplex/single-dilation-hahn-support`); $(\mathbf{No}, \mathbf{Oz})$ does not, even with set parameters. The two expansions differ exactly on the monomial section, and there is no conflict: the shears do not commute with S_2 , in line with that report's centralizer theorem (`dsup:thm:centralizer`). The shears are not exponential: by `cor:question54` of the rigidity report `docs/surreal/exponential-automorphism-rigidity` (from its `thm:no`), every exponential 1-automorphism of $\mathbf{No}$ is the identity, which is stronger than [7, Proposition 5.2] (strongly linear exponential 1-automorphisms are trivial), the result cited by 09. The shears are compatible with that report's displacement lemma: their displacement $-t^{-\eta+1}$ is infinite. Nor do they preserve the omega-map, whose compatibility questions [7, Questions 5.6, 5.7] are untouched here.

10 Surcomplex and Gaussian consequences

Identify $\mathbf{No}(i)$ with $\mathbb{C}((t^{\mathbf{No}}))$ by taking coefficientwise real and imaginary parts; the union of the two supports is well ordered. Then $\mathbf{Oz}[i] = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}} = \mathbf{Oz} + i\mathbf{Oz}$, which is not an ordered ring; conjugation is coefficientwise [04, 08, 09].

Theorem 10.1 (Gaussian stabilizer and conjugation-compatible witnesses [04, 09]). *Theorem 3.3 applies to $(k, \mathfrak{o}) = (\mathbb{C}, \mathbb{Z}[i])$ with the same condition. The real-coefficient shears and actions of sections 7 and 9 extend $\mathbb{C}$ -linearly, commute with conjugation, preserve $\mathbf{No}$ and $\mathbf{Oz}[i]$; for every set of surcomplex parameters they fix the parameters and give nondefinability of the monomials and of any valuation representative rule, even with the real axis, conjugation and every ordinary complex constant named. Their common fixed field is $\mathbb{C}$, and for every set $A \subseteq \mathbf{No}(i)$ the real elementary shears fixing A have common fixed field $\mathbb{C}((t^{V_A}))$.*

Proof. The criterion was proved for every characteristic-zero k and unital $\mathfrak{o}$, and its proof uses no order on coefficients. The functionals q_β are real, so the derivations and their exponentials commute with conjugation and restrict to the real automorphisms on $\mathbf{No}$. Use the union of the real and imaginary supports in lemma 7.1; the moved monomial is real, and lemma 9.3 holds with any nonzero complex leading coefficient. For the fixed field, the first displacement of theorem 4.2 is a product of a nonzero real functional value and a nonzero complex coefficient. The hull statement is theorem 7.4 with $k = \mathbb{C}$ [04]. $\square$

For these real-axis-preserving maps the surreal modulus is equivariant, $|\sigma(z)| = \sigma(|z|)$, since $|z|$ is the nonnegative square root of $z\bar{z}$; this is not a statement that moduli are fixed [09].

Theorem 10.2 (The real axis inside the leading-term kernel [04]). *For every set $A \subseteq \mathbf{No}(i)$ there is a strong $\mathbb{C}$ -linear 1-automorphism of $\mathbf{No}(i)$ that preserves $\mathbf{Oz}[i]$, fixes A pointwise and does not preserve $\mathbf{No}$. Hence $\mathbf{No}$ is not definable in $(\mathbf{No}(i), 0, 1, +, \cdot, \mathbf{Oz}[i])$ with parameters from any set, even with all ordinary complex constants named and the Hahn valuation ring as a predicate.*

Proof. Take β above the inner exponents of the outer supports of A , and the flow Φ_s of theorem 4.1 with $\theta = q_\beta$, $\delta = 1$ and the imaginary time $s = i$. It fixes A and is strong, $\mathbb{C}$ -linear, leading-term preserving and $\mathbf{Oz}[i]$ -preserving. With $\eta = \omega^\beta$, $\Phi_i(t^{-\eta}) = t^{-\eta} \exp(-it) = t^{-\eta} - it^{-\eta+1} - \frac{1}{2}t^{-\eta+2} + \dots$ has a nonreal coefficient, although $t^{-\eta}$ is real; all its exponents are negative. A 1-automorphism preserves every valuation, hence the valuation ring. $\square$

Phase twists preserving $\mathbf{Oz}[i]$ and moving the real axis are already in the collection (`saut:thm:phase`; `odg:def:lem:twist`, `odg:def:thm:norealaxis`); the new points are arbitrary set-parameter fixing *inside the complete leading-term kernel* and multiterm images. The real shears commute with conjugation and the imaginary-time shear does not; the two conclusions must not be combined into a claim that one automorphism commutes with conjugation and moves its fixed field [04].

Proposition 10.3 (A wild coefficient lift [08]). *There is a strong automorphism of $\mathbf{No}(i)$ that preserves $\mathbf{Oz}[i]$, fixes $\mathbb{Z}[i]$ and every Conway monomial, and sends an ordinary real number outside $\mathbb{R}$. So neither $\mathbb{R}$ nor $\mathbf{No}$ is invariant under all field automorphisms preserving $\mathbf{Oz}[i]$.*

Proof. Choose a real r transcendental over $\mathbb{Q}$ and a transcendence basis of $\mathbb{C}$ over $\overline{\mathbb{Q}}$ containing it. On the rational function field it generates, fix $\overline{\mathbb{Q}}$ and the other basis elements and send $r \mapsto r + i$ (inverse $r \mapsto r - i$), and extend to an automorphism ρ of $\mathbb{C}$ by the set-sized algebraic-closure extension theorem; the extension is onto, since its image is algebraically closed and $\mathbb{C}$ is algebraic over the base. The coefficientwise lift $\hat{\rho}$ (`odg:def:prop:coeffaut`) preserves $\mathbf{Oz}[i]$ and monomials and sends r to $r + i$. $\square$

Corollary 10.4 (Gaussian fixed field [08, 09]). *The common fixed elements in $\mathbf{No}(i)$ of all field automorphisms preserving $\mathbf{Oz}[i]$ are exactly $\mathbb{Q}$; in $\mathbf{Oz}[i]$ they are $\mathbb{Z}$.*

Proof. The shears of [theorem 10.1](#) reduce a common fixed point to $\mathbb{C}$. Every field automorphism of $\mathbb{C}$ lifts coefficientwise and preserves $\mathbf{Oz}[i]$ (`odg:def:prop:coeffaut`). A nonrational algebraic number is moved by a conjugate embedding extended to $\mathbb{C}$; a transcendental one by a translation of a transcendence basis element; rationals are fixed. Finally $\mathbb{Q} \cap \mathbf{Oz}[i] = \mathbb{Z}$. $\square$

This agrees with `saut:prop:fixedfields` of the surcomplex automorphism report, $\text{Fix}(\text{Aut } \mathbf{No}(i)) = \mathbb{Q}$, for the larger collection of all field automorphisms. The real and complex answers differ because a real automorphism preserves order and cannot move an ordinary real, whereas a complex one can unless real structure is required; the explicit witnesses of [theorem 10.1](#) fix all of $\mathbb{C}$ [09]. Three mechanisms move the real axis while preserving $\mathbf{Oz}[i]$: the phase twist of the Diophantine report (fixes $\mathbb{C}$, preserves supports), the imaginary-time shear of [theorem 10.2](#) (fixes $\mathbb{C}$ and all leading terms), and the wild lift of [proposition 10.3](#) (fixes monomials, moves $\mathbb{R}$). The real shears of [theorem 10.1](#) preserve the real axis.

11 Scope, answered questions, and further questions

11.1 What is settled, and status in other reports

For a strong infinitesimal automorphism, preservation of the integer part is exactly a convex-scale annihilation condition; the size of the residual symmetry is measured by a sharp solvability bound in finite rank and by arbitrarily large parameter-fixing actions on $\mathbf{No}$; and the omnific predicate recovers the coefficient fields and zero-cut truncation but no definable valuation cross-section [09]. An attempt to reconstruct Conway monomials by a first-order formula in the pair cannot succeed with any set of constants; a classification of strong $\mathbf{Oz}$ -automorphisms by coefficient and exponent actions alone misses the higher-rank factor $\mathcal{U}_{\mathbf{Oz}}$; and an alleged rank-one nontrivial 1-automorphism preserving a divisible-exponent integer part must violate a support hypothesis, as [example 4.12](#) shows [09].

The results bear on three questions of other reports. They are recorded here; the other reports are not edited by this merge.

- `osq:q:invisible` of `docs/surreal/set-sized-quotients-of-omnific-integers` asks, in its second clause, how much an automorphism of $\mathbf{Oz}$ can do inside Π while fixing `ct`. *Partly answered:* every automorphism of $\mathbf{Oz}$ fixes $\mathbb{R}$ and `ct` (`odg:def:thm:autreal`, [corollary 5.2](#)); the *strong* ones are exactly $\{M_{\chi,\tau}\} \ltimes \mathcal{U}_{\mathbf{Oz}}$, with $\mathcal{U}_{\mathbf{Oz}}$ classified by [theorem 3.3](#); they move monomials to infinite series, fix any set of parameters, carry $\mathbb{R}^\kappa$ and have nonsolvable stabilizers ([theorems 7.6, 7.7](#) and [9.1](#)). Nonstrong automorphisms, and the first clause of that question, remain open.
- `odg:def:q:realform` of the Diophantine report asks which natural structure on $\mathbf{Oz}[i]$ rules out the coefficient-fixing twists and determines the real omnific subring. *Negative information, not an answer:* naming the valuation ring, every complex constant, the RV sort, or requiring leading-term fixing does not suffice ([theorem 10.2](#)); and in $(\mathbf{No}(i), \mathbf{Oz}[i])$ even $\mathbb{R} \subseteq \mathbb{C}$ is not invariant ([proposition 10.3](#)).
- The surcomplex automorphism report’s research direction “effective descriptions inside the leading-term kernel” (its section `saut:sec:questions`) is *not* answered. Its $\mathbf{Oz}$ -stabilizing part is now described by coefficient functionals ([theorem 3.3](#) and [corollary 3.5](#)), with finite support certificates on the common-positive-shift class ([theorem 4.8](#)); no exhaustion or generation theorem is proved ([question 11.4](#)). Its omega-map compatibility questions are untouched.

The novelty statements of the sources were checked against the tree at the merge: no report contains a classification of $\mathbf{Oz}$ -preserving automorphisms, the rank dichotomy or set-parameter nondefinability of monomials, and `osq:q:invisible` was recorded as open. Their reconstruction and definability claims were stale (section 8).

11.2 Further questions

These questions arise from the proofs; they are not presented as named historical problems [04, 08, 09].

Question 11.1 (Strongness [04, 09]). Which ring automorphisms of $\mathbf{Oz}$ extend to strong field automorphisms of $\mathbf{No}$? Does preservation of $\mathbf{Oz}$ force strongness, or is there a nonstrong $\mathbf{Oz}$ -preserving example? Equivalently for 09: characterize the nonstrong automorphisms of $(\mathbf{No}, \mathbf{Oz})$ that commute with all set-indexed normal-form sums.

Corollary 5.2 removes coefficient motion from this question but does not settle preservation of infinite sums; nothing is inferred from the existence of nonstrong automorphisms of the ambient field [7].

Question 11.2 (Admissible logarithmic characters [04]). Give testable necessary and sufficient conditions on an additive $h : \Gamma \rightarrow \mathfrak{m}$ for $t^g \mapsto t^g \exp h(g)$ to define a strong bijective field map. Within that class the integer-part condition is exactly $a_\delta(C_\delta) = 0$.

Common positive support is one sufficient regime; the convex condition alone is not asserted to solve the global problem. This is the problem the surcomplex report describes under “what monomial prescriptions do not automatically give”.

Question 11.3 (Full relative fixed fields [04]). For a set A of parameters, determine $\text{Fix}((\mathcal{U}_{\mathbf{Oz}})_A)$. It is contained in $\mathbb{R}((t^{V_A}))$, with equality for monomial parameters (corollary 7.5); which conditions on A force equality?

Question 11.4 (Generation and exhaustion [04, 08]). How much of $\mathcal{U}_{\mathbf{Oz}}$ is generated by elementary shears and their support-controlled limits, and can an intrinsic summation convention make this precise without one common shift set for all inputs? Can all strong leading-term-fixing $\mathbf{Oz}$ -preserving automorphisms be described by a necessary and sufficient uniform support condition on their monomial phases?

No density or generation theorem is claimed; fine-topology convergence is not a substitute for a strong summation rule; the finite-dimensional phase condition of theorem 4.17 is sufficient only.

Question 11.5 (Weaker normal-form data [08]). Which normal-form data weaker than the omega-map remove the monomial nondefinability? Set-sized collections of constants do not (theorem 9.1).

Question 11.6 (Non-divisible exponent groups [merge]). For non-divisible Γ and $\mathfrak{o} = \mathbb{Z}$, is a strong k -linear 1-automorphism preserving $\mathcal{R}_{\mathbb{Z}}$ characterized by a convex-annihilator condition, as single flows are (theorem 4.10)?

This records the scope correction of remark 4.11.

Part II

Algebraic-parameter rigidity of the omnific integers

This part is manuscript 02. Its question is different from that of [part I](#): not which automorphisms preserve a support condition, but whether **Oz** admits nonconstant *algebraic* families of embeddings. It does not: the decisive property is that every nonzero element divides an element with roots of every order. It does admit formal families of all orders. The relation to [part I](#) is recorded in [section 19](#).

12 Conventions and the root-covered property

12.1 Conventions of this part

Ring homomorphisms are unital, domains are commutative integral domains, and $\mathbb{N} = \{0, 1, 2, \dots\}$; degrees of polynomials are ordinary integers. This part keeps its source's *large-monomial* convention, which is that of the other omnific reports: for a characteristic-zero field k , a unital subring $\mathfrak{o} \subseteq k$ and a nonzero divisible ordered abelian group Γ ,

$$k((X^\Gamma)) = \left\{ \sum_{g \in S} a_g X^g : S \subseteq \Gamma \text{ reverse well ordered} \right\}, \quad X^g = t^{-g},$$

so the leading exponent $\ell(x) = \max \text{supp } x$ is the largest, and $\ell(xy) = \ell(x) + \ell(y)$. The ideal $\Pi_k(\Gamma)$ consists of the series supported in $\Gamma^{>0}$ (the same set as Π in [part I](#)), and $\mathcal{R}_{\mathfrak{o}}(k, \Gamma) = \mathfrak{o} + \Pi_k(\Gamma)$ is the ring $\mathcal{R}_{\mathfrak{o}}$ of [part I](#), the Diophantine report's $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$. For surreals $X = \omega$, so **Oz** = $\mathcal{R}_{\mathbb{Z}}(\mathbb{R}, \mathbf{No})$ and **Oz**[i] = $\mathcal{R}_{\mathbb{Z}[i]}(\mathbb{C}, \mathbf{No})$. Renamings from the source: its $R_\Gamma(D, k)$ and $\mathcal{P}_\Gamma(k)$ are $\mathcal{R}_{\mathfrak{o}}(k, \Gamma)$ and $\Pi_k(\Gamma)$; its principal convex subgroup $H(a)$ is $C(a)$, the transcendence report's notation (C_δ in [part I](#); the quotient report's H_a means the opposite, the elements infinitesimal relative to a); its covering element c is z ; its bounded-scale field K_b is K_{bd} and its stages H_j are $\Gamma_{\leq j}$. Elsewhere in this part the letters R, B, A, K, F, L denote rings and fields local to each statement.

The general theorems are proved for sets. Their conclusions for **Oz**, **Oz**[i], **No** and **No**(i) use class functions and set-supported normal forms. A statement about one polynomial contains finitely many coefficients and variables, so the degree arguments apply to a class-sized coefficient domain unchanged. No proper-class scheme $\text{Spec}(\mathbf{Oz})$ and no class compactness or Noetherian theorem is used; the finite-type theorem is a theorem about sets, applied to the set-sized models of [section 18](#). Expressions involving $\text{Aut}(\mathbf{Oz})$ or ML of a class ring abbreviate statements about each automorphism or derivation; $(b, x) \mapsto D_b(x)$ is one definable class operation. For cancellation against a class-sized B , the transcendence-degree arguments involve only finite tuples of indeterminates inside one rational-function field [02].

Definition 12.1 ([02]). A domain R is *root-covered* if for every $0 \neq a \in R$ there are $b \in R$ and $z = ab \neq 0$ such that for every $n \geq 1$ some $z_n \in R$ has $z_n^n = z$. (No compatibility between the z_n is required; the terminology is local to this report.)

Fields are root-covered. The interesting examples have few units: $\mathbf{Oz}^\times = \{\pm 1\}$ and $\mathbf{Oz}[i]^\times = \{\pm 1, \pm i\}$, and in general $\mathcal{R}_{\mathfrak{o}}(k, \Gamma)^\times = \mathfrak{o}^\times$ by additivity of leading exponents (the Diophantine report's `odg:prop:units`, `odg:def:lem:units`).

Lemma 12.2 (Localization [02]). *If R is root-covered and $S \subseteq R \setminus \{0\}$ is multiplicative, then $S^{-1}R$ is root-covered.*

Proof. $(a/s)(sb) = z$, and the roots z_n remain. □

Root-covered is sufficient for the conclusions below and is not asserted necessary; failure of the monomial criterion is not used to claim failure of cancellation or stable rigidity [02].

13 The monomial-divisibility criterion

For $a > 0$ in Γ let $C(a) = \{g : |g| \leq na \text{ for some } n \geq 1\}$; a is an *order unit* if $C(a) = \Gamma$.

Theorem 13.1 (Monomial divisibility [02]; after `bst:thm:fieldcriterion`). *For $\Gamma \neq 0$ divisible and $R = \mathcal{R}_o(k, \Gamma)$ the following are equivalent: (i) Γ has no order unit; (ii) for every $0 \neq x \in R$ some monomial X^c with $c > 0$ is divisible by x in R . Then the quotient can be chosen in $\Pi_k(\Gamma)$, and R is root-covered.*

Proof. Assume (i). The support of x lies in $[0, a]$ for some $a > 0$. The inverse of x in the Hahn field has support in the subgroup generated by $\text{supp } x$, hence in $C(a)$ (write $x = \alpha X^h(1 + \varepsilon)$ and expand the geometric series). Choose $c > na$ for all n ; then $X^c x^{-1}$ has strictly positive support, lies in $\Pi_k(\Gamma)$, and $x \cdot X^c x^{-1} = X^c$. Divisibility of Γ gives $(X^{c/n})^n = X^c$. Conversely, if a is an order unit and $x = 1 + X^a$, the unique Hahn quotient $X^c/(1 + X^a) = \sum_{n \geq 1} (-1)^{n-1} X^{c-na}$ has a nonzero coefficient at a negative exponent, so it is not in R . □

This is the transcendence report’s field criterion `bst:thm:fieldcriterion` (`docs/surreal/transcendence-over-bounded-support`: the bounded-support ring is a field if and only if Γ has no order unit, by the same two arguments, the inverse support in a proper convex subgroup and the geometric series of $1 - t^u$), restricted to positive-support elements; its “order unit” is defined identically. Source 02’s audit read only the openings of the omnific reports and did not see it. The positive-support phrasing is what the rigidity theorems use.

Example 13.2 ([02]; [9]). In **No**, $\omega > n$ for all ordinary n , so $\omega^\omega/(1 + \omega) = \omega^{\omega-1} - \omega^{\omega-2} + \omega^{\omega-3} - \dots$ is an omnific integer: $1 + \omega$ divides ω^ω . With exponent group $\mathbb{Q}$ no exponent can absorb the tail. The example is L’Innocente and Mantova’s; the distinction is a larger Archimedean scale, not convergence.

Corollary 13.3 (Omnific root coverage [02]). **Oz** and **Oz**[i] are root-covered: every nonzero element divides a positive real monomial ω^c , with quotient of strictly positive support.

Proof. This is immediate from the global monomial clearing `odg:thm:fractions` of the Diophantine report, applied to x^{-1} : some ω^c makes $\omega^c x^{-1}$ purely infinite. Source 02 gives the same argument through [theorem 13.1](#), using a strict upper bound of the set $\{na : n \geq 1\}$; the coefficients’ order is not used. □

14 Finite-type domains carry no varying embedding

The results of this section concern set-sized fields and domains.

Lemma 14.1 (Unbounded roots [02]). *Let F/K be finitely generated and $u \in F^\times$. If u has n th roots in F for an unbounded set of n , then u is algebraic over K .*

Proof. Otherwise extend u to a transcendence basis $u, t_2, \dots, t_r$, with $[F : E] = d < \infty$ for $E = K(u, t_2, \dots, t_r)$. $Z^n - u$ is Eisenstein at the prime u of $K(t_2, \dots, t_r)[u]$, hence irreducible over E by Gauss's lemma, so an n th root has degree $n \leq d$. $\square$

Lemma 14.2 (Algebraic constants [02]). *If F/K is finitely generated, the relative algebraic closure K' of K in F is finite over K ; if A is an integral K -algebra with $\text{Frac } A = F$, then $C_A = A \cap K'$ is a field finite over K .*

Proof. With a transcendence basis $\mathbf{t}$ and $[F : K(\mathbf{t})] = d$, every finitely generated $L \subseteq K'$ has $[L : K] = [L(\mathbf{t}) : K(\mathbf{t})] \leq d$ (adjoining algebraically independent indeterminates preserves linear independence; $\mathbf{t}$ stays independent over L). An L of maximal degree contains every element of K' . For $0 \neq a \in C_A$ the minimal polynomial has nonzero constant term, which expresses a^{-1} as a polynomial in a over K . $\square$

Lemma 14.3 (Finitely many unit translates [02]). *Let A be a finitely generated integral K -algebra and $f \in A$ transcendental over K . Only finitely many $\lambda \in K$ make $f - \lambda$ a unit of A .*

Proof. Extend f to a transcendence basis $f, t_2, \dots, t_r$ of $\text{Frac } A$, and put $B_0 = K[f, t_2, \dots, t_r]$. Generators of A are algebraic over $\text{Frac } B_0$; clearing denominators gives $0 \neq d \in B_0$ with A integral over $B = B_0[1/d]$ in the common overring. If $(f - \lambda)^{-1} \in A$, it lies in $\text{Frac } B_0$ and is integral over the integrally closed B , so $f - \lambda$ divides a power of d in the UFD B_0 , hence divides d ; the primes $f - \lambda$ are pairwise nonassociate. The argument does not assume $B \subseteq A$. $\square$

Theorem 14.4 (Finite-type embedding rigidity [02]). *Let R be a characteristic-zero root-covered domain, K a characteristic-zero field, and A a finitely generated integral K -algebra. Every injective homomorphism $\phi : R \rightarrow A$ has image in the field $C_A = \{x \in A : x \text{ algebraic over } K\}$, finite over K , and extends uniquely to $\text{Frac } R \rightarrow C_A$. If K is relatively algebraically closed in $\text{Frac } A$, then $C_A = K$. Neither K -linearity of ϕ nor a map $K \rightarrow R$ is assumed.*

Proof. For $a \neq 0$ take $ab = z$ and $z_n^n = z$. Then $\phi(z) \neq 0$ has roots of all orders, so it is algebraic (lemma 14.1) and a unit of A (lemma 14.2); hence $\phi(a)$ is a unit. If $f = \phi(a)$ were transcendental, every $f - m = \phi(a - m)$, $m \in \mathbb{Z}$, would be a unit (as $a - m \neq 0$), contradicting lemma 14.3. So $\phi(R) \subseteq C_A$, and nonzero images are invertible there. $\square$

The theorem is nontrivial only for set-sized R , such as the Hahn models of section 18; it is not a disguised statement that a proper class cannot inject into a set. One must not replace C_A by the whole relative algebraic closure: $A = \mathbb{Q}[x, \sqrt{2}x]$ has $\text{Frac } A = \mathbb{Q}(\sqrt{2})(x)$ but $C_A = \mathbb{Q}$, since elements of A have rational constant term; no normality of A is assumed [02]. If K is relatively algebraically closed in $\text{Frac } A$, every embedding $R \rightarrow A$ has image in K , and none exists if $\text{Frac } R$ does not embed in K .

Theorem 14.5 (Pointed parameter spaces [02]). *Let R be a set-sized characteristic-zero root-covered domain, $F = \text{Frac } R$, A a finite-type integral F -algebra with an F -algebra map $\varepsilon : A \rightarrow F$. If a ring homomorphism $\psi : R \rightarrow A$ has $\varepsilon\psi$ equal to the inclusion, then $\psi(x) = x$ for all x .*

Proof. ψ is injective, so $\psi(R) \subseteq C_A$. The restriction of ε to the field C_A is injective and fixes F ; for $a \in C_A$, $a - \varepsilon(a) \in C_A$ has image 0, so $C_A = F$. $\square$

Examples are $A = F[s_1, \dots, s_r, 1/h]$ with an F -point where $h \neq 0$; relations, singularities and localizations are allowed, and relative algebraic closedness is not required. The hypothesis is one common finite-type algebra containing the images of all of R , not separate formulas in unrelated algebras. For a finite extension L/F , $L[s]$ contains the root-covered L as constants and has no F -valued augmentation when $L \neq F$ [02].

15 Polynomial and Laurent coefficient rigidity

These proofs work over arbitrary coefficient domains, need no finite generation and assume no preservation of infinite sums [02].

Theorem 15.1 (Polynomial coefficient rigidity [02]). *If R is root-covered and B a domain, every injective $\phi : R \rightarrow B[T_1, \dots, T_n]$ has image in B .*

Proof. $\deg \phi(z) = n \deg \phi(z_n)$ for all n forces $\deg \phi(z) = 0$, and $\deg \phi(a) + \deg \phi(b) = 0$. $\square$

Lemma 15.2 (Laurent units [02]). *For a domain B , the units of $B[U_1^{\pm 1}, \dots, U_q^{\pm 1}]$ are the $b\mathbf{U}^{\mathbf{e}}$ with $b \in B^\times$ and $\mathbf{e} \in \mathbb{Z}^q$.*

Proof. The widths $M_j - m_j$ of the U_j -exponents are nonnegative and additive, so a unit has a single exponent vector. $\square$

Theorem 15.3 (Laurent coefficient rigidity [02]). *If R is root-covered and B a domain, every injective $\phi : R \rightarrow B[\mathbf{U}^{\pm 1}]$ has image in B ; hence so does every injective $\phi : R \rightarrow B[\mathbf{T}, \mathbf{U}^{\pm 1}]$.*

Proof. The largest and smallest U_j -exponents of $\phi(z)$ are divisible by every n , so $\phi(z)$ is a nonzero constant, a unit over $\text{Frac } B$, and $\phi(a)$ is a Laurent monomial by lemma 15.2. If $a \neq -1$ the same holds for $\phi(a) + 1$, which forces $\phi(a)$ to be constant. For mixed targets apply theorem 15.1 over $B[\mathbf{U}^{\pm 1}]$ first. $\square$

Injectivity and a domain target are needed: $x \mapsto x + sD(x)$ into $R[s]/(s^2)$ can be nonconstant with identity specialization (section 17). No noninjective analogue for all targets is claimed [02].

Theorem 15.4 (Strong coefficient invariance [02]). *Let R be root-covered, B any commutative unital ring, and $\Phi : R[X_1, \dots, X_p, U_1^{\pm 1}, \dots, U_q^{\pm 1}] \rightarrow B[Y_1, \dots, Y_{p'}, V_1^{\pm 1}, \dots, V_{q'}^{\pm 1}]$ an isomorphism with $p + q = p' + q' < \infty$. Then $\Phi(R) = B$, $q = q'$ and $p = p'$. In particular, for $R = \mathbf{Oz}$ or $\mathbf{Oz}[i]$, every isomorphism $R[\mathbf{X}_n] \cong B[\mathbf{Y}_n]$ carries R onto B : R is strongly invariant and cancels against an arbitrary coefficient ring with the same number of variables. If B is also root-covered, the equality of total variable counts can be dropped.*

Proof. The target is a domain, so B is, and theorem 15.3 gives $\Phi(R) \subseteq B$. With $F = \text{Frac } \Phi(R) \subseteq L = \text{Frac } B \subseteq E$, $\text{trdeg}_F E = p + q = p' + q' = \text{trdeg}_L E$, so L/F is algebraic. E is a rational function field over F , which has no new algebraic constants (a nonconstant rational function is transcendental, one variable at a time), so $B \subseteq F$, and comparing coefficients gives $B \subseteq \Phi(R)$. Polynomial variables add no units, and the quotient unit groups give $\mathbb{Z}^q \cong \mathbb{Z}^{q'}$. If both rings are root-covered, apply the first step to Φ and Φ^{-1} . $\square$

Equality of total variable counts cannot be dropped for arbitrary B : $R[X_1, X_2] = B[X_2]$ with $B = R[X_1]$. The general statements do not solve the unrestricted cancellation problem for finitely generated algebras; the strong-invariance terminology and history are older [6] [02].

Theorem 15.5 (Automorphisms of polynomial and torus rings [02]). *For a root-covered R , every automorphism of $R[\mathbf{T}]$ or $R[\mathbf{U}^{\pm 1}]$ preserves R ; for polynomial extensions, restriction to R is split surjective onto $\text{Aut } R$, with kernel the R -algebra automorphisms. Every automorphism of $R[T]$ is $T \mapsto uT + b$ with $u \in R^\times$, $b \in R$, over a unique $\sigma \in \text{Aut } R$; every automorphism of $R[\mathbf{U}^{\pm 1}]$ is $U_i \mapsto u_i \prod_j U_j^{m_{ji}}$ over σ , with $u_i \in R^\times$ and $(m_{ji}) \in \text{GL}_q(\mathbb{Z})$. For $\mathbf{Oz}$, $u \in \{\pm 1\}$; for $\mathbf{Oz}[i]$, $u \in \{\pm 1, \pm i\}$.*

Proof. Apply [theorem 15.3](#) to an automorphism and its inverse; coefficientwise extension is the section. In one variable, $\deg g \deg f = 1$; units give the torus formula via [lemma 15.2](#). $\square$

The affine and torus formulas are standard once coefficients are preserved; the new omnific conclusion is that *every* ring automorphism preserves $\mathbf{Oz}$. No classification of multivariable polynomial automorphisms over R and no claim about the Jacobian conjecture follows [02].

16 Locally nilpotent derivations, coactions and gradings

A derivation D of a domain is *locally nilpotent* if every element is killed by some iterate, and $\text{ML}(A) = \bigcap_{D \in \text{LND}(A)} \ker D$. Derivations are ring derivations, hence $\mathbb{Z}$ -linear; A need not contain $\mathbb{Q}$.

Lemma 16.1 ([02]). *For a locally nilpotent D of a characteristic-zero domain, $\deg_D(a) = \max\{j : D^j a \neq 0\}$ satisfies $\deg_D(ab) = \deg_D a + \deg_D b$.*

Proof. $D^{r+s}(ab) = \binom{r+s}{r} D^r a D^s b \neq 0$, and higher iterates vanish. $\square$

Theorem 16.2 (Stable locally nilpotent rigidity [02]). *Let R be a characteristic-zero root-covered domain. Every locally nilpotent derivation of any domain containing R kills R , and $\text{ML}(R[T_1, \dots, T_n]) = R$ for every finite n . In particular $\text{ML}(\mathbf{Oz}[\mathbf{T}]) = \mathbf{Oz}$, $\text{ML}(\mathbf{Oz}[i][\mathbf{T}]) = \mathbf{Oz}[i]$, and $\mathbf{Oz}$ has no nonzero locally nilpotent derivation.*

Proof. $\deg_D(z) = m \deg_D(z_m)$ for all m forces $\deg_D z = 0$, hence $\deg_D a = 0$. The partial derivatives $\partial/\partial T_j$ are locally nilpotent with common kernel R . $\square$

Corollary 16.3 (No coactions or gradings [02]). *If R is root-covered, a homomorphism $\psi : R \rightarrow R[s_1, \dots, s_r]$ with identity specialization at $\mathbf{s} = 0$, or $\theta : R \rightarrow R[u_1^{\pm 1}, \dots, u_r^{\pm 1}]$ with identity specialization at $\mathbf{u} = 1$, is the constant inclusion; so there are no nontrivial additive-group or split-torus coactions of finite rank, and every $\mathbb{Z}^r$ -grading of R is concentrated in degree zero.*

Proof. Identity specialization makes the map injective; apply [theorems 15.1](#) and [15.3](#). A grading gives $\sum x_{\mathbf{m}} \mapsto \sum x_{\mathbf{m}} \mathbf{u}^{\mathbf{m}}$. $\square$

A Hahn expansion $\sum a_g X^g$ is not a grading: it has infinitely many terms, and the no-grading result concerns finite homogeneous decompositions. [Theorem 16.2](#) does not say that $R[\mathbf{T}]$ has no locally nilpotent derivation, and it is not a solution of a general stable-rigidity conjecture [3] [02].

17 Euler derivations and formal flows

The parameter s in this section is a new formal indeterminate, not a surreal and not a Hahn monomial.

Proposition 17.1 (Support-preserving Euler derivations [02]). *For additive $\chi : \Gamma \rightarrow (k, +)$, $D_\chi(\sum a_g X^g) = \sum \chi(g) a_g X^g$ is a k -linear derivation of $k((X^\Gamma))$ that preserves $\mathcal{R}_0(k, \Gamma)$, kills $\mathfrak{o}$ and maps $\mathcal{R}_0(k, \Gamma)$ into $\Pi_k(\Gamma)$; and $D_\chi D_\psi = D_\psi D_\chi$ for additive χ, ψ .*

Proof. The coefficient of X^h in xy is a finite sum of $a_g b_{h-g}$, and $\chi(h) = \chi(g) + \chi(h-g)$ splits it; $\chi(0) = 0$. $\square$

In the notation of [part I](#), D_χ is the shift-zero derivation $t^\gamma \mapsto \chi(-\gamma)t^\gamma$. Division by $n!$ is allowed on $\Pi_k(\Gamma)$, and $D_\chi^n(x) \in \Pi_k(\Gamma)$ for $n \geq 1$, so

$$\mathcal{E}_{\chi,s}(x) = \sum_{n \geq 0} \frac{D_\chi^n(x)}{n!} s^n \quad (17.1)$$

has coefficients in $\mathcal{R}_0(k, \Gamma)$; no division of an integer constant occurs.

Theorem 17.2 (Compatible formal flows [02]). *$\mathcal{E}_{\chi,s}$ is a unital ring homomorphism $\mathcal{R}_0(k, \Gamma) \rightarrow \mathcal{R}_0(k, \Gamma)[[s]]$ with identity specialization, satisfying $\mathcal{E}_{\chi,s} \mathcal{E}_{\chi,t} = \mathcal{E}_{\chi,s+t}$; its coefficientwise extension fixing s is an automorphism of $\mathcal{R}_0(k, \Gamma)[[s]]$ with inverse from $-s$, and it induces compatible automorphisms of $\mathcal{R}_0(k, \Gamma)[s]/(s^{N+1})$ for every N .*

Proof. $D_\chi^n(xy)/n! = \sum_r D_\chi^r(x)/r! \cdot D_\chi^{n-r}(y)/(n-r)!$, first in the Hahn field; the coefficient of $s^r t^m$ in both sides of the flow law is $D_\chi^{r+m}(x)/(r!m!)$; each coefficient uses finitely many inputs. Substituting $t = -s$ is valid since both parameters have positive order. $\square$

Theorem 17.3 (Nonalgebraization at the first jet [02]). *If Γ has no order unit and $\chi \neq 0$, the reduction of $\mathcal{E}_{\chi,s}$ modulo s^2 is nonconstant and is induced by no homomorphism $\psi : \mathcal{R}_0(k, \Gamma) \rightarrow \mathcal{R}_0(k, \Gamma)[s]$ with $\text{ev}_{s=0} \psi = \text{id}$, and D_χ is not locally nilpotent.*

Proof. Some $g > 0$ has $\chi(g) \neq 0$; then $D_\chi^n(X^g) = \chi(g)^n X^g \neq 0$. By [theorem 13.1](#) the ring is root-covered, and by [corollary 16.3](#) every such ψ is constant. $\square$

Formal automorphisms exist to *all* orders; the obstruction appears only when a single polynomial family must realize even the first-order part. A formal power series ring is not of finite type, and $R[s]/(s^2)$ is not a domain, so there is no conflict with [theorem 14.4](#) [02]. The flows involve no analytic convergence, no fine-topology limit and no identification of s with a surreal infinitesimal, and they are not an exponentiation of a derivation *inside* the Hahn field [02].

17.1 The monomial orbit field

Let $R = \mathcal{R}_0(k, \Gamma)$ be set sized, $F = \text{Frac } R$ (which contains k : $a = aX^g/X^g$) and $W = \chi(\Gamma)$, a $\mathbb{Q}$ -subspace of k . In $F((s))$ put

$$L_\chi = F\left(\frac{\mathcal{E}_{\chi,s}(X^g)}{X^g} : g > 0\right) = F(e^{ws} : w \in W), \quad e^{ws} = \sum_n w^n s^n / n!. \quad (17.2)$$

It uses only transformed monomials; it is not claimed to contain the image of every Hahn series [02].

Lemma 17.4 (Formal exponential independence [02]). *For distinct $\lambda_j \in F$ the $e^{\lambda_j s}$ are F -linearly independent; for $\mathbb{Q}$ -independent $w_1, \dots, w_r$ the $e^{w_j s}$ are algebraically independent over F .*

Proof. Comparing coefficients of s^n , $n < m$, gives a Vandermonde system. A polynomial relation is a linear relation among $e^{(\alpha \cdot \mathbf{w})s}$ with distinct exponents. $\square$

Theorem 17.5 (Monomial orbit field [02]). $\text{trdeg}_F L_\chi = \dim_{\mathbb{Q}} W$. *If $\chi \neq 0$, L_χ lies in no finitely generated extension of F inside $F((s))$, so $\mathcal{E}_{\chi,s} : R \rightarrow F[[s]]$ factors through no finite-type F -algebra with an F -algebra second map.*

Proof. A $\mathbb{Q}$ -basis (w_j) of W gives algebraically independent $e^{w_j s}$, and each $e^{w s}$ has a power equal to a monomial in finitely many of them, so L_χ is algebraic over $F(e^{w_j s})$. For $0 \neq w \in W$, $e^{w s}$ is transcendental and has roots $e^{(w/n)s}$ in L_χ , so lemma 14.1 excludes a finitely generated overfield, even when $\dim_{\mathbb{Q}} W = 1$. A factorization would put the image of R in a finite-type domain $B \subseteq F[[s]]$ with $\text{Frac } B \supseteq L_\chi$. $\square$

17.2 An omnific family of commuting directions

Theorem 17.6 (Commuting omnific derivations [02]; after `osq:prop:classder`). *For $b \in \mathbf{No}$, $D_b(\sum a_g \omega^g) = \sum q_b(g) a_g \omega^g$ is an $\mathbb{R}$ -linear derivation of $\mathbf{No}$ and, with complex coefficients, a $\mathbb{C}$ -linear derivation of $\mathbf{No}(i)$ commuting with conjugation; it preserves $\mathbf{Oz}$ and $\mathbf{Oz}[i]$. The D_b commute, and every finite family with distinct indices is linearly independent over $\mathbf{Oz}$, respectively $\mathbf{Oz}[i]$.*

Proof. Proposition 17.1 with $\chi = q_b$. For $m_c = \omega^{\omega^c} \in \mathbf{Oz}$, $D_b(m_c) = m_c$ if $b = c$ and 0 otherwise; applying $\sum_j a_j D_{b_j} = 0$ to m_{b_ℓ} gives $a_\ell m_{b_\ell} = 0$ in a domain. $\square$

These derivations, with the test elements m_c , are the quotient report's class-valued derivations `osq:prop:classder` (`docs/surreal/set-sized-quotients-of-omnific-integers`), which combine that report's sources 06, 08 and 10 and state independence over $\mathbb{R}$; the same evaluation gives independence over the omnific ring, as 02 states, and 02 adds their commutation and flows. In the t -convention of part I, $D_b(t^\gamma) = -q_b(\gamma)t^\gamma$.

Corollary 17.7 (Multi-parameter flows [02]). *For distinct $b_1, \dots, b_r$ and $R = \mathbf{Oz}$ or $\mathbf{Oz}[i]$, the map*

$$x \mapsto \sum_{\alpha \in \mathbb{N}^r} \frac{D_{b_1}^{\alpha_1} \dots D_{b_r}^{\alpha_r}(x)}{\alpha_1! \dots \alpha_r!} s_1^{\alpha_1} \dots s_r^{\alpha_r} \quad \text{into } R[[s_1, \dots, s_r]]$$

obeys the additive formal group law; its coordinate directions are independent, and no nonzero coordinate direction is induced by a polynomial family on R .

Proof. Compose one-parameter flows, which commute; independence is theorem 17.6, and the last claim is theorem 17.3 in one parameter. $\square$

Example 17.8 ([02]). For $b = 0$, $q_0(g)$ is the real constant coefficient of the exponent g : $D_0(\omega^r) = r\omega^r$ for real r , $D_0(\omega^\omega) = 0$, and $\mathcal{E}_{0,s}(\omega) = \omega + \omega s + \frac{\omega}{2}s^2 + \frac{\omega}{6}s^3 + \dots$, every coefficient in $\mathbf{Oz}$ although $\frac{1}{2}, \frac{1}{6} \notin \mathbf{Oz}$; “ ωe^s ” abbreviates this series and is not the surreal exponential. Also $D_0(\omega^\omega/(1+\omega)) = -\omega^{\omega+1}/(1+\omega)^2$, again omnific, supported at $\omega - n$ for positive ordinary n (checked coefficientwise to 29 terms by the placement dossier).

Corollary 17.9 (A continuum of independent orbit elements [02]). *For $R = \mathbf{Oz}$, $F = \mathbf{No}$ (or $R = \mathbf{Oz}[i]$, $F = \mathbf{No}(i)$), the monomial orbit field $F(e^{rs} : r \in \mathbb{R})$ of D_0 has transcendence degree $2^{\aleph_0}$ over F , in the sense of a set-sized transcendence basis with finite algebraic relations.*

Proof. $q_0(r) = r$, and $e^{rs} = \mathcal{E}_{0,s}(\omega^r)/\omega^r$ for $r > 0$. A $\mathbb{Q}$ -basis of $\mathbb{R}$, made positive, has $2^{\aleph_0}$ elements; lemma 17.4 holds over the class field F since each relation uses finitely many coefficients. $\square$

Only a lower bound is asserted for the field generated by the formal images of all Hahn series [02].

Remark 17.10 (Actual automorphisms remain abundant [02]). For additive $\chi : \mathbf{No} \rightarrow \mathbb{R}$ and ordinary $t \in \mathbb{R}$, $\tau_{\chi,t}(\sum a_g \omega^g) = \sum e^{t\chi(g)} a_g \omega^g$ is a support-preserving automorphism of $\mathbf{No}$ and $\mathbf{No}(i)$ preserving $\mathbf{Oz}$ and $\mathbf{Oz}[i]$, with $\tau_{\chi,t}\tau_{\chi,u} = \tau_{\chi,t+u}$ and positive multipliers. These character twists are established [7] and not claimed new (remark 9.5); they prevent reading the rigidity theorems as $\text{Aut}(\mathbf{Oz}) = 1$. What fails is realization of a family by a homomorphism into a polynomial or finite-type parameter algebra: with $\chi = q_0$ the twists send ω^r to $e^{tr}\omega^r$ for all real r at once, which no single polynomial parameter algebra can carry.

18 Set-sized workspaces and a fraction-field warning

Let $\Gamma = \bigoplus_{j \geq 0} \mathbb{Q}e_j$, ordered by the sign of the coefficient at the largest index, so $e_{j+1} > ne_j$ for all n : Γ is divisible with no order unit. For $(\mathfrak{o}, k) = (\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$, $R = \mathcal{R}_{\mathfrak{o}}(k, \Gamma)$ is a root-covered set (theorem 13.1), so all the results of this part apply to it in ordinary set theory. The order embedding $e_j \mapsto \omega^j$ of exponent groups induces $X^{e_j} \mapsto \omega^{\omega^j}$ and embeds R in $\mathbf{Oz}$, respectively $\mathbf{Oz}[i]$; the coordinate projections give commuting derivations already here, and replacing $\mathbb{Q}e_j$ by $\mathbb{R}e_j$ gives a set-sized realization of corollary 17.9 [02].

Let $\Gamma_{\leq j} = \bigoplus_{\ell \leq j} \mathbb{Q}e_\ell$ (convex) and $K_{\text{bd}} = \bigcup_j k((X^{\Gamma_{\leq j}}))$.

Proposition 18.1 (Bounded-scale fraction field [02]; after `bst:prop:localdensity`). *For this R , $\text{Frac } R = K_{\text{bd}} \subsetneq k((X^\Gamma))$.*

Proof. An element of R with positive leading exponent in $\Gamma_{\leq j}$ has all its exponents in $\Gamma_{\leq j}$ by convexity, so quotients lie in K_{bd} . Conversely $y \in k((X^{\Gamma_{\leq j}}))$ gives $X^{e_{j+1}}y \in R$. Finally $\sum_{j \geq 0} X^{-e_j}$ is a Hahn series in no $k((X^{\Gamma_{\leq j}}))$. $\square$

This is the transcendence report's localization proposition `bst:prop:localdensity` together with its field criterion (condition (2) of `bst:thm:fieldcriterion`) for this group. For the full class ring, $\text{Frac } \mathbf{Oz} = \mathbf{No}$ and $\text{Frac } \mathbf{Oz}[i] = \mathbf{No}(i)$ (`odg:thm:fractions`); that fact does not justify replacing K_{bd} by the full Hahn field, since a fixed set-sized group may lack an upper bound for a set of its scales even without an order unit [02].

19 Relation to Part I and the questions of this part

The two parts are complementary. Part I concerns derivations with *positive* shifts and their exponentials inside the Hahn field; this part concerns shift-zero (support-preserving) derivations and needs an external formal parameter.

- 02's question on formal directions below is partly answered by part I: the positive-shift derivations preserving $\mathbf{Oz}$ are classified exactly, by $c_\delta(C_\delta) = 0$ (theorem 3.4) and, on common positive shifts, by $a_\delta(C_\delta) = 0$ (theorem 4.7). Those with a common well-ordered set of shifts integrate *inside* $\mathbf{No}$ to automorphisms of $\mathbf{Oz}$ (theorem 4.8), so they are integrable by an honest automorphism rather than only by a formal flow; for an arbitrary contracting strong one this holds granted cited theorem 1.1 for $\Gamma = \mathbf{No}$ (remark 1.2).

The shift-zero derivations, the Hasse–Schmidt extensions of other derivations of $\mathbf{Oz}$, and derivations with no common well-ordered shift set are not covered.

- There is no conflict with [theorem 16.2](#): the shift derivations $D_{\delta,\theta}$ restrict to derivations of $\mathbf{Oz}$ but are not locally nilpotent, since $D_{\delta,\theta}^n(t^\gamma) = \theta(\gamma)^n t^{\gamma+n\delta} \neq 0$ when $\theta(\gamma) \neq 0$; and the flows Φ_s are not polynomial in s , since $\Phi_s(t^{-\eta})$ involves every power of s .
- The character twists of [remark 17.10](#) are the factor $M_{\chi,\text{id}}$ of [theorem 5.1](#).

[Theorem 15.5](#) concerns $\text{Aut}(\mathbf{Oz}[T])$, not $\text{Aut}(\mathbf{Oz})$, and does not bear on `osq:q:invisible`. The rational self-maps of $\mathbf{Oz}$ studied in the quotient report are maps of the ring given by rational functions, not ring automorphisms of $\mathbf{Oz}[T]$; the objects differ. Derivations into set-sized modules vanish (`osq:thm:derivations`), which does not conflict with [theorem 17.6](#), whose values lie in the class module Π .

Question 19.1 (Beyond the scale criterion [02]). For a nonzero divisible Γ *with* an order unit, which rings $\mathcal{R}_o(k, \Gamma)$ remain strongly invariant or stably rigid?

[Theorem 13.1](#) shows that their elements need not divide monomials; it does not show that cancellation or locally nilpotent rigidity fails.

Question 19.2 (Formal directions [02]; re-scoped). Which derivations of $\mathbf{Oz}$ preserve its purely infinite ideal, and which admit an iterative integral Hasse–Schmidt extension $\mathbf{Oz} \rightarrow \mathbf{Oz}[[s]]$? How much larger is this class than the Euler family of [theorem 17.6](#)?

Status. Partly answered for positive shifts by [part I](#), as recorded above; open otherwise. The factorial argument of [theorem 17.2](#) is a sufficient condition for the displayed family and is not asserted to classify all integrable derivations or all formal automorphisms tangent to the identity.

Question 19.3 (Reduced parameter algebras [02]). How far does coefficient rigidity extend to reduced parameter rings with several irreducible components when a prescribed specialization stays injective on the source?

The present proofs apply to domain quotients on which the map stays injective; a component can kill root-cover elements, and the dual-number example shows that nilpotent parameter rings need a different statement.

A Provenance ledger

A.1 Section by section

Section	Sources and choices
1	Structure from 09; imported results from all three automorphism sources; the Fact 4.2 recheck is the merge’s (remark 1.2) .
2	04’s one-shift lemma and coefficient polynomial; 09’s and 04’s ring decompositions; 04’s and 08’s divisible part. Units and constant products are cited from the Diophantine report, not reprinted.

Section	Sources and choices
3	The criterion is printed once. <i>Choice:</i> 09's generality (k of characteristic zero, unital $\mathfrak{o}$) with 04's import-free proof; the first implication is 09's, which removes 04's use of $\bigcap n\mathfrak{o} = 0$. 09's derivation form is printed as theorem 3.4 ; its operator form, which uses cited theorem 1.1 , as corollary 3.5 . Example 3.2 is from the placement dossier's check.
4	Shears from 04 and 09 (after <code>saut:thm:shiftflow</code>); exact displacement from 04 (08 and 09 have the same computation); 09's uniform shifts; 04's common-shift correspondence as a second, proved route; 08's divisibility-free threshold as a marked case; the translation example from 04, 08, 09 and the surcomplex report; rank dichotomy from 04 and 09 (08's for single flows); bottom subgroup from 04; phase shears from 08.
5	Factorization from 04 and 09 (credited to [8] and <code>saut:thm:decomp</code>); all pair automorphisms from 08 and 09 through <code>odg:def:thm:autreal</code> .
6	Bracket law from 04, 08, 09; filtration, rank bound, balanced brackets, mixed-parameter lemma and sharpness from 09; 04's commutator formula and 08's faithful $\mathbb{Z} \wr \mathbb{Z}$ printed as one group in 04's normalization; the all-times remark is the merge's.
7	Fixed field $\mathbb{R}$ from 04, 08, 09 (explicit choice of 08, 09); separation, hull, monomial parameters and closure from 04; $\mathbb{R}^\kappa$ and nonsolvability from 09.
8	Reconstruction and definitions of $\mathbb{Z}$ and Π from 04, 08, 09 are cited to the Diophantine report; printed extras: 08's order formula, the projections (09), floor (08), truncation and the set-sized version (09).
9	Set-parameter nondefinability from 04, 08, 09 printed once with 04's strengthening; representative rules from 09; twists, one-term class and RV from 04; contrasts added by the merge.
10	Complex criterion and conjugation-compatible witnesses from 09 and 04; real axis inside the kernel from 04; wild lift and Gaussian fixed field from 08 and 09 (09's definition of $\mathbb{C}$ is cited to the Diophantine report).
11	Status records and questions: 04's four, 08's first and third, 09's nonstrong question (merged with 04's first), and the merge's non-divisibility question.
part II	Manuscript 02 throughout, with credits to <code>bst:thm:fieldcriterion</code> , <code>bst:prop:localdensity</code> , <code>osq:prop:classder</code> and <code>odg:thm:fractions</code> ; section 19 is the merge's.

A.2 Printed once, and where the merge chose

- *The classification* (04 thm. criterion, 08 thm. threshold, 09 thms. derivation and stabilizer): [theorem 3.3](#) with 04’s proof, [theorem 3.4](#), [corollary 3.5](#), and 08’s [theorem 4.10](#) as the divisibility-free single-flow case. 04’s $a_{\sigma,\delta}$ and 09’s c_δ are kept as different functionals with equivalent conditions ([example 3.2](#)).
- *Rank triviality* (04, 09; 08 for single flows): [theorem 4.13](#). *Common fixed field* $\mathbb{R}$ (04, 08, 09): [theorem 7.2](#). *Set-parameter nondefinability* (04, 08, 09): [theorem 9.1](#).
- *The metabelian non-nilpotent group*: 04’s $\langle A_s, B_u \rangle$ and 08’s $\langle A_1, B_1 \rangle$ are the same group at $s = u = -1$; printed once as [theorems 6.8](#) and [6.10](#).
- *Strong factorization* (04, 09): [theorem 5.1](#).
- *Not reprinted* (cited to the Diophantine report): units and constant products, Pell rigidity, $\text{Frac } \mathbf{Oz} = \mathbf{No}$, the multiplier ring, the reconstruction of $\mathbb{R}$ and of the order, automorphisms fix $\mathbb{R}$, the split restriction to $\text{Aut } \mathbb{C}$, the first-order and existential definitions of $\mathbb{Z}$, Π and $\mathbb{C}$.
- *Kept as second routes*: 04’s common-shift correspondence ([theorem 4.8](#)) beside the imported one; 04’s separation lemma beside the explicit choice in [theorem 7.2](#); 09’s Pell-divisibility definition of $\Pi_{\mathbb{C}}$ is mentioned beside the Diophantine report’s.
- *Added by the merge* (tagged [merge], complete proofs): the combined statement of [theorem 3.3](#); [example 3.2](#); [remark 6.11](#); [question 11.6](#).

A.3 Corrections and stale statements

1. *Divisibility* (09): its remark that divisibility is essential is correct for $\mathcal{A}_k$ and not for $\mathcal{R}_{\mathbb{Z}}$: the translation does not preserve $\mathbb{Z} + t^{-1}\mathbb{R}[t^{-1}]$ ([remark 4.11](#)).
2. *Fact 4.2 of [7]* (09): the placement dossier doubted that it states the bijection; the merge re-read the PDF and found that it does, with only an assertion for proper classes ([remark 1.2](#)). 09’s claims $\mathcal{U}_{\mathbf{Oz}} = \text{Exp}(\mathfrak{L}_A)$ and the upper half of “exactly $r - 1$ ” in its nonsolvability theorem rest on that assertion and are stated so ([corollary 3.5](#) and [theorem 7.7](#)).
3. *Stale novelty* (04, 08, 09): their reconstruction, first-order recovery of $\mathbb{R}$ and definitions of Π were written at pins `fb5c4b53` and `befe739b`, before the Diophantine report’s reconstruction section was placed; they are cited, not claimed ([section 8](#)). 04’s remark that its reconstruction is not first order is superseded by `odg:def:cor:internal`. 08’s definition of Π has the structure of `odg:cor:definablect`, which was in the tree at 08’s pin.
4. *08’s Question 2* is answered ([section 8](#)) and dropped.
5. *Credits added*: `saut:thm:shiftflow` (08 and 09 did not credit it), `saut:thm:decomp` (04 and 09 credited only [8]), the rigidity report’s `thm:no` and `cor:question54` (09 cited only [7, Proposition 5.2]), `bst:thm:fieldcriterion`, `bst:prop:localdensity`, `osq:prop:classder` and `odg:thm:fractions` (for 02); and the surcomplex report’s $T_1(t) = t/(1 - t)$ is noted not to preserve $\mathbf{Oz}$ ([example 4.12](#)).
6. *Repository statements checked true*: at both pins the catalogue listed 51 reports (04, 08); the fixed-shift flow is `saut:thm:shiftflow` (04); phase twists preserve $\mathbf{Oz}[i]$ and move the real axis (04); the exponential-rigidity question is answered in its report (04); Pell rigidity and a definition of $\mathbb{Z}$ were in the Diophantine report (08); $\text{Frac } \mathbf{Oz} = \mathbf{No}$ was

there (09, 02); no report contained an **Oz**-automorphism classification (04, 08, 09). 04 reports two date fields for [7] v3 (23 April and 24 August 2026); it is cited as version 3 of 23 April 2026, as elsewhere in the collection.

A.4 Symbols renamed from the sources

$\mathcal{J}$ (08, 09) and $\mathcal{I}$ (04) are Π ; B (09), $\mathcal{A}$ (04), $\mathcal{B}_k$ (08) are $\mathcal{A}_k$; A_D (09), O_D (04), A_k (08), $R_\Gamma(D, k)$ (02) are $\mathcal{R}_\circ$ or $\mathcal{R}_\circ(k, \Gamma)$, with the coefficient ring D renamed $\circ$ because D names derivations; $\mathcal{P}_\Gamma(k)$ (02) is $\Pi_k(\Gamma)$; H_δ (08) and $H(a)$ (02) are C_δ and, in [part II](#), $C(a)$; $\mathcal{U}_{A_D}$ (09) and $\mathcal{U}_D(G)$ (04) are $\mathcal{U}_{\mathcal{R}_\circ}(\Gamma)$; $D_{\varphi, \delta}$ (08) is $D_{\delta, -\varphi}$; q_η (09), $[\omega^\beta] \cdot$ (04), c_a (08) and χ_b (02) are q_β ; 09's $H = \omega^\eta$ and H_j are $\eta = \omega^\beta$ and η_j ; 08's A_1, B_1, u, E are $A_{-1}, B_{-1}, u_0, \exp$; 02's covering element c is z , its K_b and H_j are K_{bd} and $\Gamma_{\leq j}$. 08's ω -convention statements are translated to the t -convention ([section 1.5](#)); [part II](#) keeps 02's large-monomial convention ([section 12.1](#)).

B Non-claims, by source

No source claims to solve a named conjecture, and none claims Lean verification, peer review or historical priority; the finite suites check identities only. The report keeps every limitation stated by each source.

Source 09.

1. The classification is only within strong, coefficient-fixing, leading-term-preserving automorphisms; it does not assert that an abstract automorphism preserving $\mathcal{R}_\circ$ is strong, and nonstrong pair automorphisms are not classified (the definability recovery applies to them; the logarithmic classification does not).
2. Coefficient functionals come from an existing strong derivation; arbitrary functions satisfying the vanishing condition are not asserted to assemble into an operator.
3. The shears are plain-field automorphisms and are not claimed to preserve the Gonshor exponential.
4. No set or NBG class of class-function graphs is formed; actions are encoded by single class relations; supports are sets.
5. The exponential–logarithm correspondence and the factorization pattern are imported, not reproved or claimed; the explicit many-shift constructions avoid the class version of the import.
6. No new factorization theorem for omnific integers is claimed; [9] addresses a different problem.
7. **Oz** is not identified with a model of full ordinary arithmetic.
8. Truncation at a named monomial is not a uniform definable choice of boundary.
9. Nondefinability is relative to the field-and-integer-part language; Conway normal forms remain set-theoretically defined.
10. Hahn-sum formulas are not claims of convergence in the fine topology.
11. Modulus equivariance is not a claim that moduli are fixed.
12. The repository comparison was bounded and targeted; search coverage is not a proof of novelty.

13. The exact-arithmetic program checks finite identities only; it does not decide well-ordering, prove the correspondence, verify class-theoretic statements or certify novelty; the formalization order is a proposal, not implemented modules.

Source 04.

1. It does not classify abstract (nonstrong) automorphisms of **Oz**; the fraction-field extension of an abstract automorphism is not known to be strong.
2. It imposes no exponential, omega-map, birthday or simplicity structure, and does not revisit the collection's exponential-rigidity question as if unanswered.
3. The criterion classifies existing maps; it is not a construction for arbitrary logarithmic characters, and the convex condition alone is not asserted to solve global admissibility.
4. The relative fixed field is computed for the elementary subgroup only; the full stabilizer only for monomial parameters; the Hahn hull is not asserted to be definable closure.
5. Class maps are treated individually; no class of all automorphisms is formed.
6. Nothing is claimed after naming a proper class of parameters.
7. Bare nondefinability of coefficient-one monomials already follows from character twists; the nondefinability statement alone is not advertised as a new mechanism.
8. $\text{Frac } \mathbf{Oz} = \mathbf{No}$ is a full-class statement; a set-sized Hahn field need not be the fraction field of its integer-part ring.
9. Its intrinsic coefficient invariance was not claimed to be first order (now superseded by the Diophantine report).
10. Real and imaginary shear times preserve different structures; its Gaussian result is not the first surcomplex automorphism moving the real axis.
11. No generation or density theorem; fine-topology convergence is not strong summation.
12. The fixed-shift exponential, the general monomial/kernel decomposition and the exponential-derivation correspondence are prior work.
13. Its finite checks are not proofs and do not implement arbitrary normal forms; no Lean file, no repository checkout or build; historical priority unverified.

Source 08.

1. The threshold covers the elementary family only; rank-one rescalings and characters preserve the integer part; not every leading-term-preserving automorphism is claimed to be the exponential of one fixed-shift derivation.
2. Finite-dimensional phase images are a sufficient condition only.
3. No first-order Gaussian reconstruction was claimed (one now exists, [section 8](#)).
4. The automorphisms are not asserted to commute with the surreal exponential, the Berarducci-Mantova derivation or the omega-map; the wreath product does not contradict rigidity for those structures.
5. Parameter avoidance fails for set-sized fields with the whole field named; a universe-relative collection is not identified with the proper class.
6. No classification of the full stabilizer, and no solution of additional-structure problems.

7. The reconstruction is not a homomorphism $\mathbf{Oz} \rightarrow \mathbb{R}$.
8. The nondefinability is not a statement about adding an arbitrary class predicate, nor about automorphisms required to commute with the omega-map.
9. Strong summation is not topological convergence.
10. Pell rigidity, the existence of nontrivial Hahn automorphisms, the derivation–exponential construction and the monomial-character viewpoint are not claimed new; some elementary lemmas may be known; priority is provisional.
11. Its questions are not named historical problems.
12. No Lean; not refereed; the checks are regression tests, not proofs of well-ordering, parameter avoidance or definability.

Source 02.

1. Root-covered is sufficient, not claimed necessary; failure of the monomial criterion is not used to claim failure of cancellation or rigidity.
2. The finite-type theorems are theorems about sets; no $\text{Spec}(\mathbf{Oz})$ and no class compactness; “ $\text{Aut}(\mathbf{Oz})$ ” and ML of class rings are shorthand; the finite-type theorem is not a statement that a class cannot inject into a set.
3. It does not classify $\text{Aut}(\mathbf{Oz})$, all derivations, or multivariable polynomial automorphisms, and makes no Jacobian-conjecture claim.
4. Not every homomorphism into every target is constant: injectivity and domain targets are required; no noninjective analogue is given.
5. It does not claim that $R[\mathbf{T}]$ has no locally nilpotent derivation; a Hahn expansion is not a grading.
6. The orbit-field formula concerns monomials; only a lower bound holds for the full formal image.
7. Formal flows involve no analytic convergence and no fine-topology limits, s is not a surreal, and they are not exponentiation inside a Hahn field.
8. The cancellation theorem does not solve the unrestricted cancellation problem; $\text{ML}(R[\mathbf{T}]) = R$ is not a solution of a general stable-rigidity conjecture.
9. Not new: character twists [7], strong-invariance terminology [6], the Makar–Limanov framework [3], Euler derivations, Vandermonde arguments, and the example of [9].
10. No named factorization conjecture is claimed solved; the literature search is not a priority certificate.
11. The questions are not presented as published open problems; no Lean, no peer review; the checks are finite identities.

Counts: 13 non-claims from 09, 13 from 04, 12 from 08, 11 from 02.

C Finite checks

Four suites ship with the report, byte-identical to their delivery, under the prefixes of their manuscripts. They check finite identities in exact arithmetic; they do not prove Hahn summability, class-sized statements, definability or novelty. They were rerun on copies for this merge with Python 3.14.4 and SymPy 1.14.0, with the recorded results reproduced.

Script and record	Content and behaviour
code/09-omnific-preserving-verify_finite_identities.py; data/09-omnific-preserving-verification_output.txt	Standard library. Product rule and bracket on finite Laurent polynomials, balanced brackets for ranks 2 to 8, mixed group-commutator coefficients with square-zero parameters, shear iterates, exponential laws, the rank-one escape, Pell growth. Twelve PASS lines and “ALL CHECKS PASSED”. Prints only.
code/04-preserving-automorphisms-verification.py; data/04-preserving-automorphisms-verification_report.json	SymPy 1.14.0 (data/04-preserving-automorphisms-requirements.txt). 7,062 assertions in 17 groups: exponential group law, bracket formula, coefficient polynomial, first corrections, commutator coefficients in lexicographic $\mathbb{Q}^3$. <i>Writes</i> its report to --output, by default verification_report.json in the working directory.
code/08-integer-part-symmetries-verify.py; data/08-integer-part-symmetries-verification.json	Standard library. 9,469 checks in 15 categories: derivation rule, falling-factorial iterates, flow composition, bracket, Vandermonde determinants, the semidirect-product law, commutator phases. <i>Always writes verification.json</i> beside itself.
code/02-parameter-rigidity-verify.py; data/02-parameter-rigidity-verification.json	Standard library. 7,041 assertions in 13 families: Euler product rule, factorial Leibniz, truncated flow composition, commutation, geometric remainders, Vandermonde determinants, binomial identity. Writes verification.json beside itself unless --output is given.

The placement dossiers also recorded independent checks (281 for the automorphism cluster, including [example 3.2](#), the commutator formula for symbolic s, u , the leading terms of [theorem 6.8](#) for $n \leq 4$, the conjugation lemma and the failure example of [remark 4.11](#); and for 02 the quotient-rule identity of [example 17.8](#) to 29 terms); these scripts are not shipped.

References

- [1] V. Bagayoko, L. S. Krapp, S. Kuhlmann, D. Panazzolo and M. Serra, *Automorphisms and derivations on algebras endowed with formal infinite sums*, [arXiv:2403.05827v2](#), September 22, 2025; especially Theorems 3.11 and 3.13.
- [2] S. Camacho, *Truncation in Hahn fields is undecidable and wild*, [arXiv:1706.03722](#), 2017.
- [3] M. Chitayat and D. Daigle, *On the rigidity of certain Pham–Brieskorn rings*, J. Algebra **550** (2020), 290–308; [arXiv:1907.13259](#).
- [4] J. H. Conway, *On Numbers and Games*, Academic Press, 1976; second edition, A K Peters, 2001.
- [5] H. Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986.
- [6] Y. Hirano, *On the uniqueness of rings of coefficients in skew polynomial rings*, Publ. Math. Debrecen **54** (1999), 489–495.
- [7] E. Kaplan, L. S. Krapp and M. Serra, *Decomposing the automorphism group of the surreal numbers*, [arXiv:2509.22374v3](#), April 23, 2026; especially Fact 4.2, Proposition 5.2 and Questions 5.4, 5.6, 5.7.
- [8] S. Kuhlmann and M. Serra, *The automorphism group of a valued field of generalised formal power series*, J. Algebra **605** (2022), 339–376; [arXiv:2107.03362](#).
- [9] S. L’Innocente and V. Mantova, *A factorisation theory for generalised power series and omnific integers*, Adv. Math. **442** (2024), 109513; [arXiv:1710.07304v5](#).

Reports of the collection are cited by directory and label: the omnific Diophantine report `docs/surreal/omnific-diophantine-geometry (odg:)`, the quotient report `docs/surreal/set-sized-quotients-of-omnific-integers (osq:)`, the surcomplex automorphism report `docs/surcomplex/surcomplex-field-automorphisms (saut:)`, the rigidity report `docs/surreal/exponential-automorphism-rigidity (bare labels)`, the transcendence report `docs/surreal/transcendence-over-bounded-support (bst:)` and the single-dilation report `docs/surcomplex/single-dilation-hahn-support (dsup:)`.